Sardar Vallabhbhai National Intitute of Technology

Department of Physics

Dissertation Report

Integrated Masters in Physics

Astrophysical Impacts on The Primordial Black Holes

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Academic Year 2021–2022

ABSTRACT

The earliest works on the Primordial Black Holes, which are the central theme of this report, or PBHs date back to 1966. It was Carr and Hawking's seminal paper in 1971 that laid the theoretical foundation of the PBH formation mechanism. PBHs first got attention from the non-cosmology community in 1975 proposed that PBHs could constitute a fraction or even entire amount of the dark matter content of the universe. Throughout the 1980s till today, several models of inflation have been constructed that could produce a large enough (of the order $10^{-2} - 10^{-4}$ perturbations in the curvature power spectrum (30 to 40 e-folds before the end of inflation), which collapse into PBHs upon horizon re-entry. This report aims to shed a light on the events aftermath of the PBH formation, driven by the astrophysics of the surroundings of the PBHs, that could affect the initial population and/or mass function of the PBHs. In the last section, we also describe the bounds on the present abundance of the PBHs established from various observations.

Contents

1 Introduction to FRW Cosmology

Before Edwin Hubble made the revolutionary discovery that most of the galaxies are receding away from the milky way (in other words, the spectra of most of the galaxies were redshifted due to the Doppler effect), it was a common notion amongst the cosmologists that the universe was static i.e. the physical scale of the universe did not depend on the time at which it was measured. Einstein too was a firm believer in a static universe. Shortly after he formulated the General Theory of Relativity in 1915, Einstein began searching for the exact solutions to his field equations

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{1}$$

where, G is the gravitational constant, same as the one in Newtonian gravity, $R_{\mu\nu}$ is the symmetric Ricci tensor, R is the trace of the Ricci tensor (also known as the Ricci scalar), $T_{\mu\nu}$ is the energy-momentum tensor of the system being considered and rest of the terms are the usual constants. He tried to apply these equations to understand the geometry of the universe by considering it as a single and isolated system. Einstein assumed that the behaviour of the universe was static and consisted of a uniformly distributed matter. To his disappointment, the equations (1) could not give a static solution for either zero or non-zero spacetime curvature. In an attempt to counter the attractive dynamics due to the gravity of the matter distribution, Einstein introduced a new term Λ, to the equation (1) that would act as a repulsive term. The modified Einstein equations now looked like

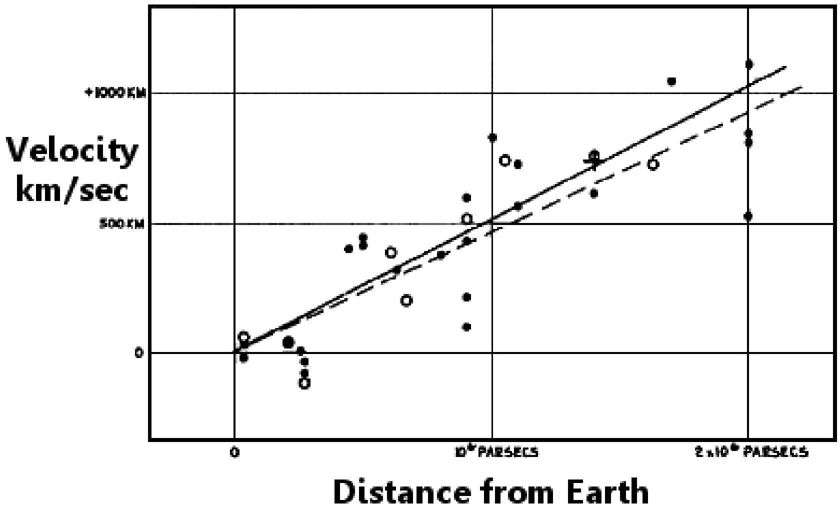

Figure 1: Hubble diagram that shows the velocity-distance relation for 46 different galaxies. Although the overall diagram was a scatter plot, Hubble had boldly drawn a straight line, indicating that the recession velocity of the galaxies directly depends on their distance from earth.[1]

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{2}$$

It is easy to check that Λ, widely known as the Cosmological Constant, has the dimension of the inverse of the square of the length. Shortly after Einstein published the static universe solution of his equations, several physicists such as Alexander Friedmann, Georges Lemaître and Willem de Sitter independently came up with their solutions to the Einstein field equations. However, these solutions represented the universe as evolving with time, unlike Einstein's model. In 1929, when Hubble discovered the direct relationship between the redshift in the spectra of 46 galaxies with their distance from earth [1], it became clear that the universe is not static as Einstein thought it to be. Eventually, Einstein discarded his model of the static universe. It is also claimed that Einstein considered the introduction of the Cosmological Constant as 'the biggest blunder of his life' [4]. We will now summarize the results of Alexander Friedmann that explain the geometry of an evolving universe.

We first list three assumptions that will simplify the computation of the field equations

1. The cosmological principle is valid [1],
2. Universe is filled with a perfect fluid,
3. The spatial scale of the universe depends on the cosmic time.

the Friedmann–Lemaître–Robertson–Walker (FLRW) metric takes the following form on the basis of assumptions 1 and 2

$$ds^2 = -dt^2 + a^2(t)d\Sigma^2 \tag{3}$$

The parameter $a(t)$, which is one of the most important physical quantities in cosmology, is called the scale factor. The main reason for dividing the four-dimensional spacetime into spatial and temporal components is because the three-metric $d\Sigma^2$ is a maximally symmetric space-metric [2] and therefore we can apply

[1] At large scales, the universe stays the same no matter where you are or in which direction you look (Homogeneous and Isotropic)

[2] A maximally symmetric space contains the highest possible number of symmetries

the direct expression for maximally symmetric spaces. The expression for $(R^3_{\alpha\beta\gamma\lambda})$ for the three-metric of a maximally symmetric space is well-known [2]

$$R^3_{\alpha\beta\gamma\lambda} = k(f_{\alpha\lambda} f_{\beta\lambda} - f_{\alpha\lambda} f_{\beta\gamma}) \tag{4}$$

where, f_{ij} is the FRW three-metric divided by the scale factor. The trace of equation (4) gives the Ricci tensor for a maximally symmetric three metric

$$R^3_{\beta\lambda} = 2k f_{\beta\lambda} \tag{5}$$

Now, a maximally symmetric three-metric possesses three symmetries. Hence, it is nothing but a spherically symmetric metric. So, the Ricci tensor for a spherically symmetric metric (or the Schwarzschild metric) is compared with (5) and the following expression for the three-metric is obtained

$$d\Sigma^2 = \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \tag{6}$$

Here, k is nothing but a scaled form of Ricci scalar that denotes the geometrical structure of the universe as a whole. It can take three distinct numerical values namely +1, 0 and -1. The three different values of k correspond to three different shapes of the universe. If $k = -1$, the universe is said to be negatively curved or a hyperbolic geometry and if $k = 1$, the universe has a positive curvature or a spherical geometry. The case $k = 0$ corresponds to a flat or Euclidean universe. Our current perception of cosmology hints at a flat universe (with astonishing accuracy!) [5].

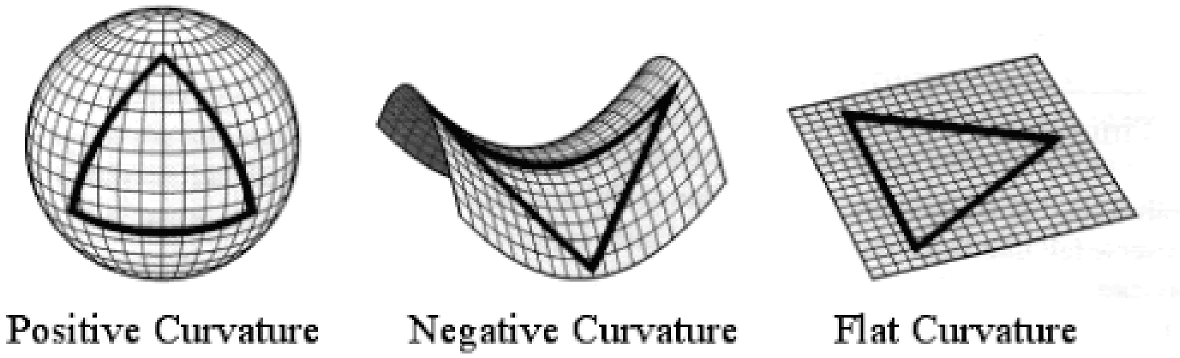

Figure 2: Geometry of the universe for different values of k [102]

Ultimately, the FRW metric can be represented in its most generic form as

$$ds^2 = dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2\left(d\theta^2 + \sin^2\theta d\phi^2\right)\right] \tag{7}$$

Now that we have got the metric, the last task is to use the Einstein equations (1) to find out how the scale factor changes with the energy content of the universe. We follow the method of [2] and start by deriving an expression for the energy-momentum tensor (T_{ij})

$$T_{ij} = (\rho + p)v_i v_j + p g_{ij} \tag{8}$$

where, v_i is the four-velocity of the fluid, p and ρ being the pressure and energy density terms respectively. Our assumption 2 (that the universe consists of a perfect fluid) makes the task much simpler. The definition demands perfect fluid to be isotropic, without any turbulence, heat conduction or viscosity. This implies that all the spatial components of the four-velocity of the fluid vanish in its comoving frame i.e. the fluid is at rest. Raising the index of (8) after substituting

$v_i = (1, 0, 0, 0)$, we get

$$T^i_j = \begin{bmatrix} -\rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{bmatrix} \tag{9}$$

The energy conservation equation in tensor notation is

$$\Delta_i T^i_0 = 0 \tag{10}$$

because the T^i_0th term represents the energy density and the usual derivative has to be replaced by a covariant derivative in curved spacetime. Expanding (10) and simultaneously using the equation of state $p = \gamma\rho$, the energy conservation equation becomes

$$\frac{\dot{\rho}}{\rho(1+\gamma)} = -\frac{3\dot{a}}{a} \tag{11}$$

For the left hand side of (1), the required non-zero terms of the Ricci scalar are

$$R_{00} = \frac{-3\ddot{a}}{a} \qquad R_{11} = \frac{2\kappa + a\ddot{a} + 2\dot{a}^2}{-kr^2 + 1} \tag{12}$$

In addition, the $\{22\}$th and $\{33\}$ terms are also non-zero. However, due to the maximally symmetric nature of the spatial three-metric, the Einstein equations will give identical results for $\{11\}$, $\{22\}$th and $\{33\}$terms.

Finally, we substitute the equations (8) (considering the perfect fluid assumption) and (12) into the Einstein equations. After simplification, the $\{00\}$th term of becomes

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = -\frac{k}{a^2} + \frac{8\pi G}{3}\rho \tag{13}$$

whereas the $\{11\}$th term becomes

$$\frac{3}{4\pi G}\frac{\ddot{a}}{a} = -(\rho + 3p) \tag{14}$$

Equations (13) and (14) set up the foundation of modern cosmology and are jointly known as the *Friedmann equations.* The Hubble parameter H determines the age as well as the span of the observable universe. Interestingly, the right-hand side of (13) is said to be representing the total energy density of the universe if the first term on the right is considered as the energy contribution due to the curvature of the universe. Note that equation (14) can be derived from (13) if the exact form of energy conservation equation (11) is known.

2 Limitations of the Standard Big Bang Model

2.1 The Flatness Problem

Before addressing this problem, let us first concentrate on some background calculation. We define a ratio Ω such that

$$\Omega_t = \frac{\rho_t}{\rho_c} \quad ; \quad \rho_t = \rho_m + \rho_r + \rho_\Lambda \tag{15}$$

also,

$$\begin{aligned} \Omega_t > 1 \quad &\Longleftrightarrow \quad \text{Positive curvature} \\ \Omega_t = 1 \quad &\Longleftrightarrow \quad \text{Zero curvature} \\ \Omega_t < 1 \quad &\Longleftrightarrow \quad \text{Negative curvature} \end{aligned}$$

where, ρ_t is the total energy density of the universe, ρ_m, ρ_r and ρ_Λ are the energy densities corresponding to the matter, radiation and the Cosmological Constant respectively. We also have ρ_c, known as the critical energy density, which is nothing but the magnitude of the energy density required to keep the universe flat. Now, rearranging the first Friedmann equation (13)

$$\rho_t a^2 - \rho_c^2 a^2 = \frac{3k}{8\pi G} \tag{16}$$

To obtain the expression for ρ_c, just put $k = 0$ in (13) and ρ_t would become ρ_c

$$\rho_c = \frac{3}{8\pi G} H^2 \tag{17}$$

Some more rearrangement to (16) gives

$$\rho a^2(1-\Omega^{-1}) = \frac{3k}{8\pi G} \tag{18}$$

The right side of the equation (18) being a constant term implies that the left side has to remain constant as the universe evolves. The scale factor in the left-hand side of (18) has a varying relationship with the cosmic time that depends on the equation of state of the universe. However, it increases with time overall. Simultaneously, the energy density term decreases in (18) with the cosmic time (again this "decrease" varies with the instantaneous equation of state of the universe). However, the rate of decrease of energy density is much (for example, $\rho_t \propto a^{-4}$ during the radiation dominated era and $\rho_t \propto a^3$ in the matter-dominated era) faster than the rate of increase of the scale factor and so the term ρa^2 decreases with time. The ref. [6] estimates that from a little time after the big bang till today, this product term has decreased by an order of 10^{60}. To compensate for this reduction. the $(1-\Omega^{-1})$ term has to increase in a similar order at the same time. This means that if the value of Ω will differ even slightly from unity, then the expansion would drive away from very far from 1. But our current results from the observations [5] predict that the current value of Ω is extremely close to one. Therefore, to achieve the current value of Ω, the initial had to be extremely fine-tuned. This is known as the Flatness problem. Looking from the perspective of the standard Big Bang model, a flat universe is nothing less than a miracle!

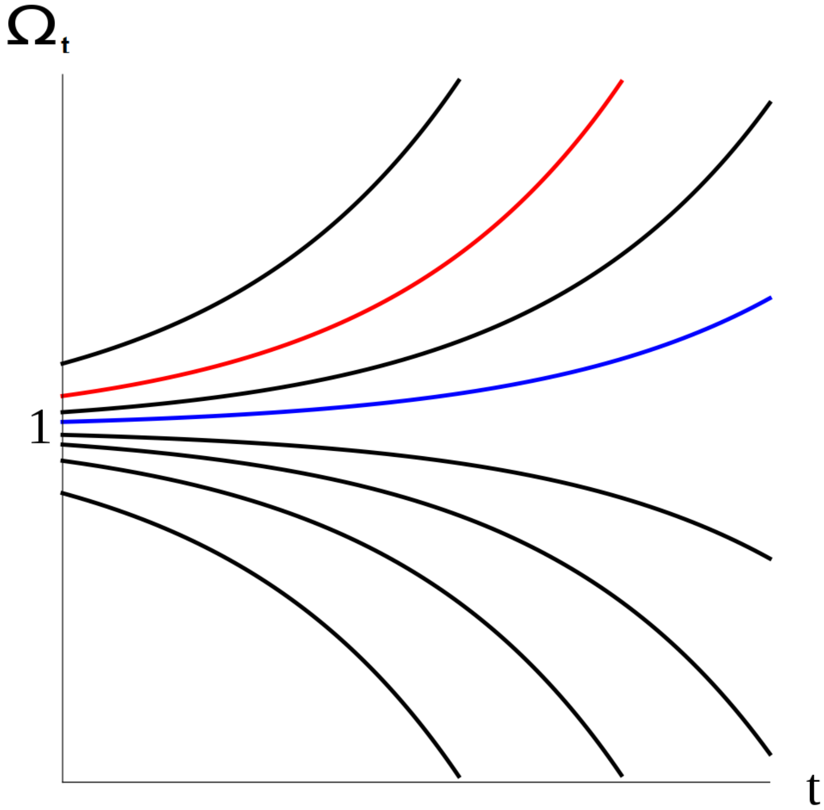

Figure 3: The graph portrays how the fractional energy density Ω_t evolves according to its different initial values. It is obvious that even a tiny deviation of the initial value of Ω_t would have driven it far away from 1 rapidly [103]

2.2 The Horizon Problem

This problem exists due to the finite speed of light in the vacuum. We calculate two different spatial scales to identify the horizon problem, distance travelled by the photon since the last scattering (r_1) and the current horizon size (r_2). Assuming that the light travelled radially and without any gravitational deviation, the total distance travelled by a photon between a time interval $\Delta t = t_f - t_i$ is

$$r_1 = \int_{\Delta t} \frac{dt}{a(t)}. \tag{19}$$

Note that this distance is expressed in a comoving reference frame. To get the physical distance travelled by the photon, equation (19) has to be multiplied by the scale factor. Equation (19) upon integrating gives [2]

$$r_1 = 2H_0^{-1}(\sqrt{a_f} - \sqrt{a_i}) \tag{20}$$

where, H_0 is the magnitude of the Hubble parameter measured today. Now, in order to find the distance travelled by photon right from recombination till present, the scale factors in the equation (20) become $a_f = 1$ and $a_i = a_{rc}$.

$$r_1 = 2H_0^{-1}(\sqrt{1} - \sqrt{a_{rc}}) \quad \approx \quad 2H_0^{-1} \tag{21}$$

The physical distance travelled by the photon would simply be

$$r_1^p = a_{rc}\ r_1 = \frac{1}{1 + z_{rc}} r_1 \approx 1.8 \times 10^{-2} H_0^{-1} \tag{22}$$

We know that the current size of the physical horizon is nothing but

$$r_2^p = H_0^{-1} \tag{23}$$

Comparing equations (22) and (23), we can clearly say that photons emitted at the time of recombination have not got enough time to travel the distance equal to the current physical horizon size. Hence, there is no way that the two widely separated points in the sky could have causally connected. However, the results of PLANCK [8] suggest that the CMB emitted from two different ends of the sky has the (almost) same temperature. In fact, this amazing uniformity in CMB temperature is observed across the entire sky. Without any causal connection in the past, it is impossible for the present CMB sky to be so homogeneous. This is known as the Horizon problem in the standard Big Bang model.

3 Inflation

The theory of cosmic inflation was proposed in 1979 by Alan Guth, who was originally looking for a solution to the non-existence of magnetic monopoles in the present universe. Theories of particle physics predict the production of extremely heavy magnetic monopoles in the very early universe when the temperature was higher than the grand unification phase transition temperature. Guth argued that this problem could be solved provided an "Inflationary phase" is added to the standard big bang model as an extension. This inflationary phase, possibly triggered by a scalar field, witnessed an extremely rapid expansion of the universe for a very short period (between 10^{-36} seconds to 10^{-32} seconds after the big bang). During this phase, the overall equation of the state of the universe behaved as if it were dominated by the cosmological constant i.e. the scale factor increased exponentially with time. Hence,

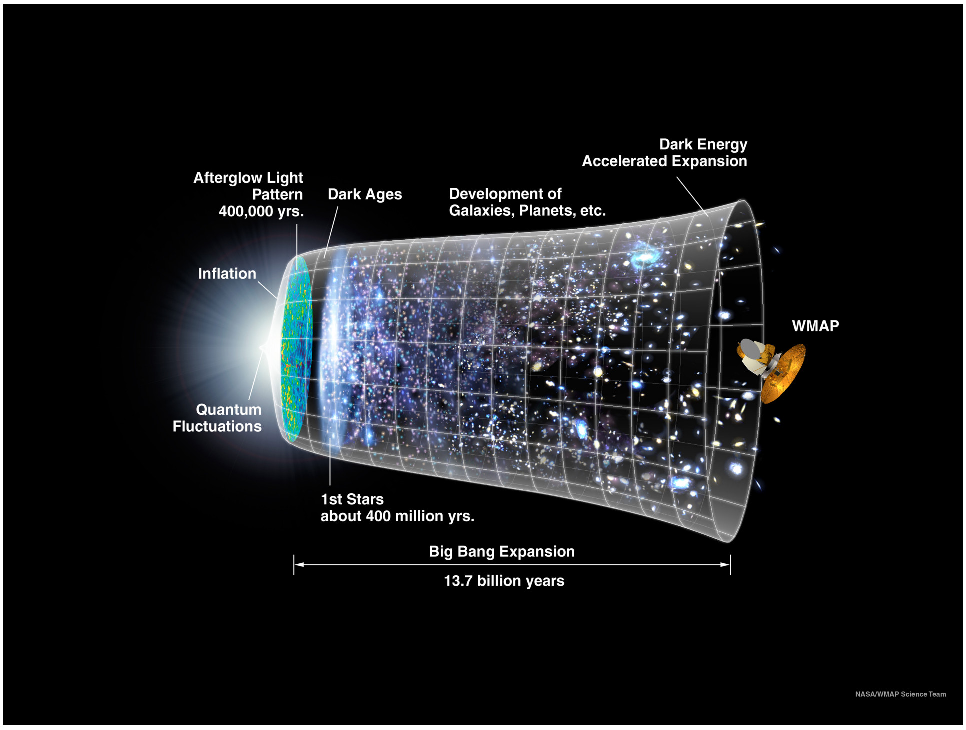

Figure 4: Modified timeline of the universe after adding the inflation package to the standard Big Bang cosmology. As shown in the figure, the inflationary era witnessed an (exponential) accelerated expansion of the universe that lasted only for a short period and then radiation dominated era took over [104].

$$p = -\rho$$

and

$$a(t) = a_0 e^{\alpha t}$$

during inflation. The last equation directly implies that the Hubble parameter remains constant throughout the inflationary phase. As a result of this rapid expansion, the magnetic monopoles got diluted as the universe evolved and hence they are not observed today.

The time period of inflation is very short so, it is not convenient to study the evolution of the universe during inflation as a function of time. Instead, a different parameter N, which is called the number of e-folds, defined as

$$\frac{a_f(t)}{a_i(t)} = e^N \tag{24}$$

is used. Here $a_i(t)$ and $a_f(t)$ are the magnitudes of the scale factor at the beginning ($t = t_i$ and end ($t = t_f$) of inflation respectively. One of the most primitive approaches to studying inflation is a single scalar field (also called inflaton field) model. The basic idea is that a scalar field [9] drives the inflation as it slowly rolls down from the maxima of the potential until it reaches the bottom of the potential suggesting the end of the inflation. After the inflation ends, the radiation dominated era takes over. Let us revisit some basic calculations for a slow-rolling inflaton field ϕ in the influence of a potential $V(\phi)$. The Lagrangian of the field is given by [2]

$$L = V(\phi) + \frac{1}{2} g^{\alpha\beta} \Delta_\alpha \phi \Delta_\beta \phi \tag{25}$$

and the corresponding energy-momentum tensor is

$$T_{\alpha\beta} = g_{\alpha\beta} \left[\frac{1}{2} g^{\lambda\gamma} \Delta_\lambda \phi \Delta_\gamma \phi - V(\phi) \right] + \Delta_\alpha \phi \Delta_\beta \phi \tag{26}$$

Comparing equations (26) and (8), we can easily obtain the following expressions

$$p(\phi) = \frac{1}{2}\dot{\phi}^2 - V(\phi) \quad \& \quad \rho(\phi) = \frac{1}{2}\dot{\phi}^2 + V(\phi) \tag{27}$$

As already mentioned, total energy of the scalar field is the only contribution to the total energy of the universe during inflation. Hence, equation (13) becomes

$$H_\phi^2 = -\frac{k}{a^2} + \frac{1}{3m_P^2}\left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right) \tag{28}$$

where m_P is the Planck mass. The behaviour of the scalar field is studied by the Klein-Gordon equation. The assumption of the cosmological principle leads to the following expression for the Klein-Gordon equation for an inflaton field

$$\ddot{\phi} = -\left(3H_\phi\dot{\phi} + V'(\phi)\right) \tag{29}$$

where the superscripts dot and prime denote the derivatives with respect to time and ϕ respectively. The first term on the right-hand side can be regarded as a friction term due to the expanding background. If the inflation has to last for a sufficient number of e-folds, the inflaton field must roll very slowly as shown in the figure 5.

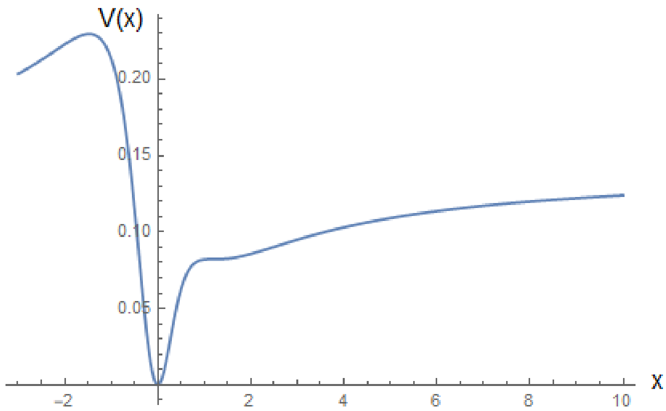

Figure 5: A scalar field potential for a toy inflaton field x. The slope of the potential is very small for $x \gtrsim 1$, indicating a slow roll phase [9].

This implies that the dynamic term or "kinetic energy" term of the scalar field has to be extremely smaller than the potential term

$$\frac{1}{2}\dot{\phi}^2 << V(\phi) \tag{30}$$

Equation (30) is the necessary condition for any slow-rolling inflationary model. The approximate version of equations (28) and (29) in case of a slow-rolling field is

$$H_\phi^2 \approx -\frac{k}{a^2} + \frac{1}{3m_P^2}V(\phi) \quad \& \quad H_\phi\dot{\phi} \approx -V'(\phi)/3 \tag{31}$$

To regulate rate of slow roll we define two parameters

$$2\epsilon(\phi) = m_P^2\left(\frac{V'(\phi)}{V(\phi)}\right)^2 \quad \& \quad \eta(\phi) = m_P^2\left(\frac{V''(\phi)}{V(\phi)}\right)^2 \tag{32}$$

known as the slow rolling parameters. In terms of these parameters, the necessary condition for slow-rolling approximation is

$$\epsilon << 1 \quad \& \quad \eta << 1 \tag{33}$$

The moment one of these parameters reaches a value of the order of unity, the slow-rolling approximation has to be discarded because the kinetic energy term of the potential can no longer be ignored. Either of the conditions $\epsilon = 1$ or $\eta = 1$ (whichever becomes one first) signify the end of inflation.

Although the magnetic monopole problem was the sole motivation for introducing inflation, surprisingly, inflation could successfully give answers to the two major problems of the standard big bang model explained in the previous section. Let us summarize how inflation solved the Flatness problem and the Horizon problem.

3.1 Solutions to the Flatness problem and the horizon problem

Let us first rewrite equation (18) for the moment just before the beginning of inflation

$$H_\phi^2 a^2 (1 - \Omega^{-1}) = \frac{3k}{8\pi G} \tag{34}$$

As already said, the value of H_ϕ remains frozen through the inflation so the Hubble parameter in the above equation can be regarded as a constant. Moreover, the right-hand side of equation (34) remains constant. This gives

$$(1 - \Omega^{-1}) \propto \frac{1}{a^2} \tag{35}$$

We know that the scale factor increases exponentially during the inflationary phase. So, the term inside the braces rapidly converges to zero no matter how small or large Ω was before the beginning of inflation. Hence, regardless of its initial value, the fractional energy density of the universe reaches extremely close to unity at the end of inflation. This solves the Flatness problem. Note that inflation should last for a sufficient amount of time to ensure a flat universe.

On account of an exponential expansion in the early universe, the actual size of the physical horizon is much larger than the (23), which is predicted by the standard big bang model. It can be easily calculated that the horizon problem can be solved provided that the epoch of inflation lasts for at least 60 e folds.

Of course, a single scalar field model is only one of the approaches to generating an inflationary phase. A plethora of sophisticated models of inflation have been proposed, for example, Models with a sharp step in the potential, that can be sketched by a discontinuity in the second derivative of the inflaton potential [12], inflationary models with a curvaton-like spectator field that acquires isocurvature fluctuations through the piling up of quantum fluctuations on different scales[13] which get converted into curvature perturbations later on, when this field dominates the energy density of the universe (provoking a temporary stage of early matter domination), Models in which an auxiliary field triggers a waterfall transition at the end of inflation, like at the end of hybrid inflation, for example, [14] and even multi-field models, where the auxiliary field triggers a second-order phase transition, for example, [15].

4 Primordial Black Holes

The concept of a stellar black hole has been studied for more than a century since the discovery of the first-ever analytical solution of the Einstein equations [16] in 1916. It is known that when a massive enough star reaches the end stages of its life cycle, it runs out of fuel required for the nuclear fusion to continue. Eventually, the radiation pressure needed to sustain the self-gravity of the star declines and the star's gravitational pull takes over. If the star is massive enough [17], even the degeneracy pressure due to fermions is not enough to resist its gravity. The moment the star's radius becomes less than its Schwarzschild radius, it collapses into a singularity, more commonly referred to as a black hole.

Primordial Black Holes or PBHs, as the name suggests, are the black holes formed during the very early history of the universe. The concept of the PBHs was first introduced by Zeldovich and Novikov in 1966 [19]. PBHs were formed a fraction of seconds after the big bang and there was not enough mass in the universe for the structure formation to take place. So, PBHs could not be formed from the mechanism by which stellar-mass black holes are formed. A number of formation mechanisms have been proposed for the PBHs, for example, the collapse of cosmic loops [23], bubble collisions [24] and the collapse of overdense regions [19]. Throughout this report, we will be only concerned with the last mechanism, which was originally proposed by Zeldovich and Novikov.

Various models of Inflation could produce relatively large fluctuations on a range of scales that could be either large or narrow but, in any case, smaller than those observed with Cosmic Microwave Background (CMB), Large Scale Structure (LSS) and Lyman-alpha forest surveys. PBH generation requires a much larger amplitude in the scalar power spectrum which cannot be achieved at CMB or LSS scales [25]. After inflation, the amplified scales are larger than the Hubble radius. They contribute to spatial fluctuations in the curvature and energy density fields, with a stochastic distribution featuring peaks and troughs on various scales. It is useful to think of these fields smoothed over sub-Hubble scales at any given time. When a given scale crosses the Hubble radius, gravitational collapse starts around overdense regions of size given by this scale. The rare regions in which the overdensities are the highest are the first to collapse. If they are sufficiently overdense, their collapse leads to the formation of a black hole.

Recently, a lot of works have formulated the mechanism of PBH formation both analytically and numerically, for example, [27, 28]. First, people have studied how the statistics of the primordial curvature perturbation field $\zeta(\vec{x})$ can be used to characterise the maxima of the density field $\delta(\vec{x})$. These maxima are usually approximated as spherically symmetric. They can be described by their profile $\delta(r)$ and their abundance as a function of their amplitude. Second, people have used analytical methods [29, 30] and numerical simulations [31, 32] to compute the threshold value δ_c above which a given overdensity collapses to form a black hole. They showed that the numerical value of this threshold depends on various assumptions, in particular, on the profile $\delta(r)$ of the overdensity. Third, people have used theoretical frameworks in order to relate the PBH abundance Ω_{PBH} and the PBH mass distribution $f(M_{PBH})$ to the primordial spectrum $\mathcal{P}_\zeta(k)$ generated by each inflationary model. These theoretical frameworks include threshold statistics (that is, the Press-Schechter theory) and different versions of peak theory and excursion set theory. Finally, people have scrutinized various observational bounds on the density of black holes of a given mass to constrain first $f(M_{PBH})$ and then $\mathcal{P}_\zeta(k)$.

There are numerous reasons why PBHs have been getting so much attention lately. For instance, PBHs are regarded as one of the potential candidates for non-baryonic dark matter. One of the most interesting properties of the PBHs is their vast mass range. Theoretically, PBHs could be as light as 10^{-5}g or as heavy as 10^{37}g [22]. Although present constraints on PBHs have ruled out the possibility of them constituting a significant fraction of dark matter, we still have a few interesting "unconstrained" mass windows for PBHs [26]. Moreover, PBHs could have also aided in the formation of structures during the matter-dominated era [33]. PBHs that was massive enough to have survived the Hawking evaporation might also have provided the seeds for the supermassive black holes (SMBHs) that reside in the centre of most of the galaxies today [34].

5 Accretion

Accretion refers to the spiralling of surrounding matter and radiation onto a central massive and compact object (such as a Primordial Black Hole) or an extended object (a Dark Matter halo, for example) as a result of its gravity. Accretion is one of the most important properties of a black hole and must be taken into account to fully understand the astrophysics of the black hole. The earliest works on accretion onto point mass objects were pioneered by Hermann Bondi, Fred Hoyle, Raymond Lyttleton et al [35, 36, 37]. This section aims to revisit the accretion model for the black holes formed during the radiation dominated era. However, first, we will summarize the two earliest accretion models viz. Hoyle-Lyttleton accretion and Bondi-Hoyle accretion will help us set up a platform to study the accretion onto PBHs. Our calculations are based on the reference [38].

5.1 Hoyle-Lyttleton Accretion

The main difference in both the accretion model lies in the assumptions made to compute the accretion rate. Hoyle and Lyttleton [35] start by assuming an infinite cloud of gas and a massive object performing a steady motion through it.

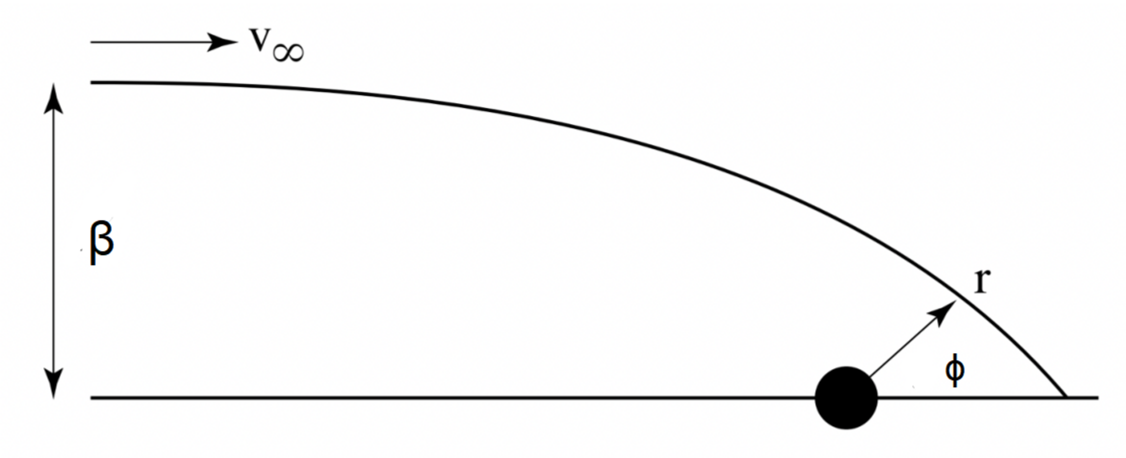

Figure 6: A schematic of Hoyle-Lyttleton accretion [38].

Figure 6 shows an infalling streamline approaching the massive object from infinity with a velocity v_∞ and impact parameter β. It has been also assumed that the effect of pressure is negligible on the trajectory, which implies that the streamline traces a ballistic orbit. Now, applying the Euler-Lagrangian equation to the system shown in the above figure, we get

$$\ddot{r} - r\dot{\phi}^2 + \frac{GM}{r^2} = 0 \tag{36}$$

here r, ϕ is the radius of the trajectory and angle between the reference horizontal axis and trajectory respectively and the dot above the variables denotes its derivative with respect to time. The conservation of angular momentum using Newtonian mechanics gives

$$r^2\dot{\phi}^2 - \beta v_\infty = 0 \tag{37}$$

After re-calibrating, Equation (36) takes the following form,

$$\frac{d^2\alpha}{d\phi^2} + \alpha - \frac{GM}{l^2} \tag{38}$$

where $u = \frac{1}{r}$ and $h = \beta v_\infty$. This is an ordinary differential equation whose general solution is given is $\alpha = A\cos\phi + B\sin\phi + C$. The arbitrary constants A, B can be obtained from the boundary conditions. Substituting the general solution into Equation (38), we get $C = \frac{GM}{l^2}$. The two boundary conditions required to find the explicit expression for A, B are

1. $\phi \to \pi \qquad \Rightarrow \qquad \alpha \to 0$
2. $\phi \to \pi \qquad \Rightarrow \qquad \dot{r} \to -v_\infty$

Applying these boundary conditions to the Equation (38) and using the expression for C, we get

$$A = \frac{GM}{l^2} \qquad \& \qquad B = -\frac{v_\infty}{l} \tag{39}$$

and hence,

$$\alpha = \frac{GM}{l^2}\cos\phi - \frac{v_\infty}{l}\sin\phi + \frac{GM}{l^2} \tag{40}$$

Now when the streamline reaches the horizontal reference axis i.e. the $\phi = 0$ axis, only the radial component of its velocity remains non zero because the force is along the line of action so their cross product (which is the net torque) will be zero. Hence $\dot{\phi} = 0$, $\dot{r} = v_\infty$ and the instantaneous value of the streamline radius can be obtained from Equation (40)

$$\alpha = r^{-1} = \frac{2GM}{l^2} \tag{41}$$

Unless the streamline is bound to the central object, it cannot accrete. Hence, a necessary condition for this is that the total energy of the system must be negative

$$E = K.E. + P.E. < 0 \qquad \Rightarrow \qquad \frac{GM}{r} > \frac{1}{2}v_\infty^2 \tag{42}$$

from this we can define a critical value of the impact parameter β_{HL}, also called the Hoyle-Lyttleton radius.

$$\beta < \beta_{HL} = \frac{2GM}{v_\infty^2} \tag{43}$$

The exact value of the Hoyle-Lyttleton radius holds quite a significance because only that matter will be accreted whose impact parameter is less than β_{HL}. We can now find out the total amount of mass being accreted onto the central object, called the Hoyle-Lyttleton accretion rate

$$\dot{M_{HL}} = \pi \beta_{HL} v_\infty \rho_\infty = \frac{4\pi G^2 M^2 \rho_\infty}{v_\infty^3} \tag{44}$$

Due to the simplified nature of Hoyle-Lyttleton accretion, the analytical solutions to the accretion parameters are quite straightforward. The expressions for density, velocity components and streamline radius have been computed in the Reference [40] and, are given as

$$\rho = \frac{\rho_\infty \beta^2}{r \sin\phi (2\beta - \sin\phi)} \tag{45}$$

$$v_\phi = \frac{v_\infty \beta}{r} \tag{46}$$

$$v_r = -\sqrt{\frac{2GM}{r} + v_\infty^2 - \frac{v_\infty^2 \beta^2}{r^2}} \tag{47}$$

$$r = \frac{v_\infty^2 \beta^2}{GM(1+\cos\phi) + \beta v_\infty^2 \sin\phi} \tag{48}$$

Except for the density equation, which can be derived from the continuity equation of a gas in a steady state, the other three equations are the solutions of the equations discussed above.

5.2 Bondi-Hoyle Accretion

Hoyle-Lyttleton's analysis had assumed the accretion column to be infinitely thin and of infinite density on the horizontal axis $\phi = 0$. This contradicts the assumption of a ballistic trajectory. Hence, Bondi and Hoyle improved this analysis by including an accretion column of finite width, as shown in Figure 7.

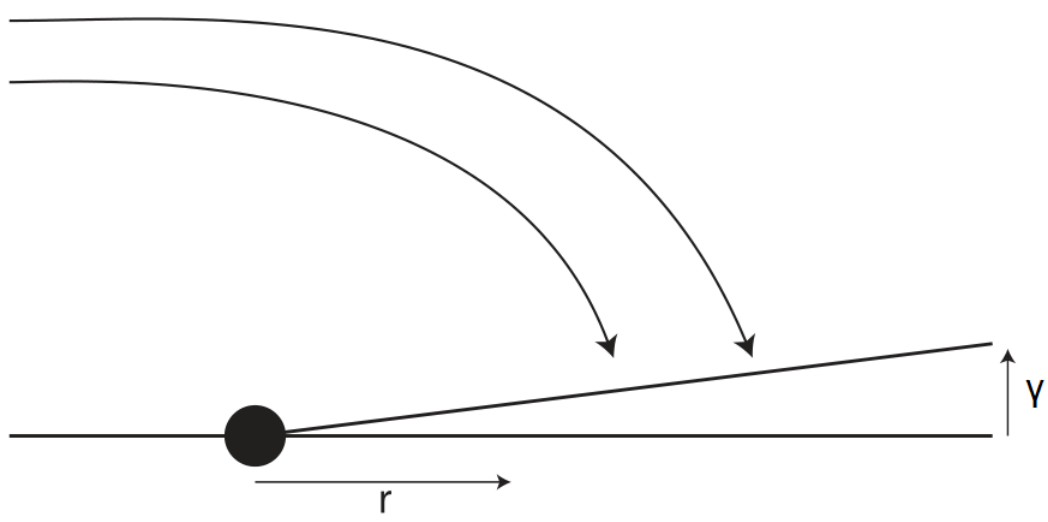

Figure 7: A schematic of the Bondi-Hoyle accretion [38].

Equation (41) suggests that the radial distance of streamline from the central object when it reaches $\phi = 0$ axis is

$$r = \frac{v_\infty^2 \beta^2}{2GM} \tag{49}$$

Now, the mass flux for an infinitesimal distance element r to $r + dr$ is

$$(2\pi\beta d\beta v_\infty)\rho_\infty = \zeta dr \tag{50}$$

where, $\zeta = \frac{2\pi GM\rho_\infty}{v_\infty}$. For the same interval, we can easily find the transverse momentum flux by multiplying the mass flux with the transverse velocity per unit wake area

$$\zeta \cdot v_\phi(\phi = 0) \cdot \frac{1}{2\pi\Gamma} \tag{51}$$

This momentum flux is nothing but the pressure in the wake (P). We get the following expression for P by using the previous orbit equation

$$P = \frac{\zeta}{2\pi\Gamma}\sqrt{\frac{2GM}{r}} \tag{52}$$

which gives longitudinal pressure force

$$d(\pi\Gamma^2 P) = \zeta\sqrt{\frac{GM}{2}}d\left(\frac{\Gamma}{\sqrt{r}}\right) \tag{53}$$

we can also calculate the mass of the wake per unit length of the wake from the accretion rate equation (44) by considering approximate infall time $\approx \frac{r}{v_\infty}$

$$m \approx \zeta\frac{GM}{v_\infty} \tag{54}$$

hence, the gravitational force per unit length

$$F = \frac{G^2M^2}{v_\infty^3} \cdot \frac{dr}{r^2}\zeta \tag{55}$$

By assuming wake length to be negligible as compared to the radial distance ($\Gamma << r$), a conical in shape ($\frac{d\gamma}{\Gamma} = \frac{dr}{r}$) and r to be of the order $\sim \frac{GM}{v_\infty^2}$, it turns out that the gravitational force in Equation (55) is much larger than the pressure force term Equation (53). Therefore, we can safely ignore the pressure effects in the wake. Assuming u as the average wake velocity, the mass flux per unit length ζ can be used to express the mass conservation and momentum conservation equations respectively

$$\frac{d}{dt}(mu) - \zeta = 0 \tag{56}$$

$$\frac{d}{dr}(mu^2) - \zeta v_\infty + \frac{GMm}{r^2} = 0 \tag{57}$$

It is better to analyze the conservation equations in their dimensionless form. Hence, by introducing dimensionless variables

$$m = \frac{\zeta GM}{v_\infty^3}\chi \tag{58}$$

$$v = v_\infty V \tag{59}$$

$$r = \frac{GM}{v_\infty^2}R \tag{60}$$

the equations (56) and (57) take the following form

$$\frac{d}{dR}(\chi V) - 1 = 0 \tag{61}$$

$$\frac{d}{dR}(\chi V^2) + \chi R^{-2} - 1 = 0 \tag{62}$$

Integrating Equation (61), we get

$$\chi V = R - k \tag{63}$$

where the constant k is the stagnation point because the dimensionless velocity changes (V) sign when $R = k$. This implies that accretion will occur only when the $R < k$ condition is met. It is now clear that k determines the rate of accretion for a wake. Equation (62) can be rewritten using Equation (63) such that

$$V\frac{dV}{dR} + \frac{1}{R^2} - \frac{V(1-V)}{R-k} \tag{64}$$

A careful analysis of Equation (64) can help to extract useful information without solving the equation. We redefine the boundary conditions discussed previously for the dimensionless variables

1. $R \to \infty \quad \Rightarrow \quad V \to 1$
2. $R = k \quad \Rightarrow \quad V = 0$ (This is the stagnation point)
3. $\frac{dV}{dR} > 0$ This condition ensures that there are no abrupt changes in the flow of the wake.

The third condition puts a significant constraint on the allowed values of k whereas, the first two conditions are quite general and satisfied by all the values of k. Now, we modify Equation (64) by substituting $\Sigma = \frac{R}{k}$

$$V\frac{dV}{d\Sigma} + \frac{1}{k\Sigma^2} - \frac{V(V-1)}{\Sigma - 1} = 0 \tag{65}$$

assuming that the derivative term i.e the first term of the above equation is zero. This gives the following exact solution for the dimensionless velocity parameter V

$$V = \frac{1}{2} \pm \sqrt{\frac{1}{4} - \frac{1}{k\Sigma^2}(x-1)} \tag{66}$$

As velocity is a real and measurable quantity, the term inside the root has to be greater than zero. This requires computing another quadratic equation, this time solving for Σ

$$\Sigma = \frac{2}{k}(1 \pm \sqrt{1-k}) \tag{67}$$

The approximate value of k is computed using numerical methods and depends on the equation of the state of the streamline that is being accreted. The analysis of Bondi and Hoyle found out that the wake becomes unstable to perturbations for $k > 2$ and hence, the axial symmetry gets distorted. Note that $k = 2$ corresponds to the Hoyle-Lyttleton accretion solution. The Bondi radius is defined as [37, 39]

$$R_B = \frac{GM}{c_s^2(R_B)} \tag{68}$$

which is nothing but a spatial boundary that differentiates the type of accretion flow. The flow is "subsonic" and uniformly dense outside the Bondi radius. On the other hand, inside the Bondi radius, the gas flow becomes "supersonic" and the density no longer remains uniform. Inspired by the similarities between the Equations (44) and (68), Bondi proposed the following formula

$$\frac{dM}{dt} = \frac{2\pi G^2 M^2 \rho_\infty}{(c_\infty^2 + v_\infty^2)^{3/2}} \tag{69}$$

Equation (69) is most commonly referred to as the *Bondi-Hoyle* accretion rate and this result will further be exploited to analyze the accretion onto PBHs in the coming sections.

6 Accretion In The Early Era

As already mentioned, PBHs could constitute a significant (or even entire) fraction of dark matter, at least for a certain mass range [25]. To fully understand the present observational constraints on the abundance of the PBHs, the parameter $f(M_{PBH}) = \frac{\Omega_{PBH}}{\Omega_{DM}}$ is used. However, various astrophysical events such as accretion, clustering, Hawking radiation and binary mergers could significantly change the PBH fraction and eventually, the present observational bounds on the PBH abundance. For example, if PBHs are modelled to be clustered in large spatial volumes rather than being isolated, it is possible to avoid the microlensing constraints [44]. Similarly, PBH accretion and merger could also alter the PBH bounds significantly. In the next subsection, we will see that the "amount" of accretion onto the PBHs is quite significant. We will also see that if the PBH dark matter fraction is less than unity ($f_{PBH} < 1$), accretion onto PBH increases even more due to the presence of a dark matter halo surrounding the PBHs.

We first explore a simple case of accretion onto the isolated PBHs by ignoring the effects of PBH clustering and PBH binary formation. We assume that the spin of the PBHs is negligible so that the spherical symmetry around the PBHs remains intact and the analysis of Bondi [37] is applicable.

6.1 Importance Of PBH Accretion

Here, we briefly revisit the results of [41] to understand the growth PBH mass due to accretion in the radiation as well as matter-dominated era. First, we lay down the assumptions taken for arriving at the results

1. The accreting material possesses a non-zero angular momentum so that the

radial infall of the material is avoided. As a result, the material does not fall into the PBH fully, rather it forms a halo around the PBH,

2. Throughout the history of PBHs, the accretion of only dark matter is considered as the fraction of baryonic matter very small compared to the dark matter fraction and accretion due to radiation is not significant in the radiation dominated era [22].

3. The accreting material in the surrounding is following the Hubble expansion.

4. The velocity is negligible as compared to the expanding FRW background. Hence, PBHs are considered to be stationary.

5. Any kind of clustering of the PBHs or binary formation is disregarded. PBHs are considered isolated.

Accretion Estimate In The Radiation Dominated Era

We start by writing the differential equation for a system of PBH surrounded by a shell of dark matter with a mean separation R

$$m\ddot{R} + \frac{GMm}{R^2} + \frac{mR}{4t^2} = 0 \tag{70}$$

where M and m are the PBH mass and mean shell mass respectively and the dot represents the derivative with respect to time. In the absence of the central mass, the differential equation (70) reduces to an unperturbed second-order equation

$$\ddot{r} + \frac{mR}{4t^2} = 0 \tag{71}$$

Assuming that the solution of this equation follows a power law

$$R_0(t) = R_i \left(\frac{t}{t_i}\right)^{\alpha} \tag{72}$$

and substituting it into Equation (71), the value of α turns out to be 0.5. The unperturbed solution now reads

$$R_0 = \frac{R_i}{t_i^{1/2}} \tag{73}$$

Adding PBH into the picture is equivalent to adding a small perturbation to the solution (73) i.e. $R = R_0 + \delta R$ and substituting this in (70), we get

$$\delta\ddot{R} + \frac{\delta R}{4t^2} + \frac{GMt_i}{tR_i^2} = 0 \tag{74}$$

where we have neglected the non-linear terms in $\frac{\delta R}{R}$. To simplify this expression, we define $y = \frac{\delta R}{R_i}$ and $T = \frac{t}{t_i}$. Hence, Equation (74) reduces to

$$\ddot{y} + \frac{y}{4T^2} + \frac{\eta}{T} = 0 \tag{75}$$

Here,

$$\eta = \frac{GMt_i^2}{R_i^3} \approx \left(\frac{\delta\rho}{\rho}\right)_{t_i} \tag{76}$$

The solution to this equation is

$$x(\eta) = 4\eta(T^{1/2} - T) = \frac{\delta R}{R_i}(T) \tag{77}$$

Now, we define another important parameter known as the collapse time (t_c), which is nothing but the time taken by the dark matter shell to merge to the PBH.

In other words, the radius of the dark matter shell becomes zero at the collapse time. Using the condition $R = R_0 + \delta R = 0$, we get the following expression for the collapse time

$$t_c = t_i \, T_c = t_i \left(\frac{1+4\eta}{\eta}\right)^2 \tag{78}$$

To estimate the amount of accretion onto the PBHs during the radiation dominated era, we take t_i as an arbitrary time in the early radiation era and t_f to be the time at the radiation-matter equality. Throughout the radiation dominated era, the redshift time relation is expressed as $t \propto 1/(1+z)$. Hence, setting $t_f = t_c$, we get

$$\frac{t_c}{t_i} = \left(\frac{1+4\eta_{min}}{4\eta_{min}}\right)^2 \approx 10^7 \tag{79}$$

where, η_{min} implies that we are considering the maximum possible magnitude of the accretion. From Equation (76), we get

$$\frac{M_{acc,max}}{M} = 4\pi R_{i,max}^3 \rho_{m,i} = \frac{4\pi}{3} \frac{G t_i^2}{\eta_{min}} \rho_{m,0} (1+z_i)^3 \tag{80}$$

The subscript 0 implies that the present value of the quantity is being used. Rearranging and substituting values of the numerical terms, we arrive at

$$\frac{M_{acc,max}}{M} \approx 3 \tag{81}$$

Here M_{acc} is the mass of the PBH attributed to the accretion. It turns out that the fractional increase in the PBH mass during the radiation dominated era is of order unity.

Accretion Estimate In The Matter Dominated Era

Note that, as the universe enters the matter-dominated era, the contribution of the cosmological constant (Λ) becomes significant. Hence, the differential equation governing the PBH and dark matter shell dynamics in the matter-dominated era is given by [45]

$$\ddot{R} + \frac{G\mu_i}{r^2} - \frac{\Lambda r}{3} = 0 \tag{82}$$

where, μ is the total mass of the region surrounded by the shell of dark matter of initial radius R_i

$$\mu_i = \frac{4\pi R_i^3 \rho_{m,i}}{3} + M \tag{83}$$

As usual, M is the PBH mass and $\rho_{m,i}$ is the initial background matter density. The above equation (82)is then solved numerically after defining the initial conditions and the accreted mass onto the PBH is given by [41]

$$\frac{M_{acc}}{M} = \left(\frac{1 - \rho_{BH,i} - \rho_{\Lambda,i}}{\theta_i + \rho_{BH,i} + \rho_{\Lambda,i}}\right) \tag{84}$$

The parameter θ_i is the initial density contrast

$$\theta = \frac{3\mu_i}{4\pi R_i^3 \rho_i} - 1 \tag{85}$$

Due to the negative pressure of the Cosmological constant, it works as an opposing force that prevents the shells of dark matter to accrete onto the PBH. Therefore, the shells located at a considerable distance from the PBH will not accrete and keep on following the background Hubble expansion. In other words, the shells outside a certain radius no longer remain bound to the PBH regime. This radius is given by

$$R_\Lambda^3 = \frac{M}{2\pi\rho_{m,i}}\left(\frac{2\Omega_{\Lambda,i}}{\Omega_{m,i}}\right) \tag{86}$$

The total mass inside this radius can be easily found by multiplying the total volume encompassed by the last bound radius to the initial background matter density $\Omega_{m,i}$. Some substitution eventually gives

$$\frac{M_\Lambda}{M} = \left(\frac{8(1-\Omega_{\Lambda,i}}{54\Omega_{\Lambda,i}}\right)^{1/3} \tag{87}$$

where M_Λ is the total mass of the last bound volumed sphere. Time at radiation-matter equality is assumed to be the beginning of the accretion. The above ratio $\frac{M_\Lambda}{M}$ could go well beyond the order 10^3. However, in reality, more than one PBHs are accreting the surrounding matter simultaneously and eventually, they would start interacting with each other and the assumption of isolated PBH fails. So, the accretion in the matter era can increase the PBH mass by order 2, which is quite large and may affect the present PBH bound tremendously.

6.2 Accretion Onto Isolated PBHs

Knowing that accretion could contribute significantly to the increase in the initial PBH mass, we now proceed to briefly discuss the derivation of basic equations for the accretion onto PBHs. We follow the earliest approach described by Ricotti et al [42]. Again, we start by stating the assumptions

1. The masses of PBHs are smaller than 10^5 solar masses. For the PBHs more

massive than $10^5 M_\odot$ the spherical accretion formulas of Bondi do not apply, as will be seen later.

2. In the early universe (when the redshift is greater than 100), the matter is coupled to the radiation (CMB) on account of Compton scattering and acts as a densely packed fluid surrounding the PBHs. This fluid resists the accretion mechanism and is hence, included in the accretion calculations as a viscosity term.

3. At the time of formation, the size of the PBH horizon is of the order of the physical horizon so, the effect of Hubble expansion on the astrophysics of PBHs cannot be ignored. The expanding background drags away the accreting matter from the PBH and thus contributes to the viscosity term.

4. Two separate cases of accretion will be considered viz. accretion onto a point mass and accretion onto a point mass surrounded by a spherical dark matter halo. The density profile of the halo is assumed to be following a power law.

The radiation pressure due to the CMB photons opposes the motion of the fluid element thereby, producing a restoring force per unit mass. The expression for this restoring acceleration is given by [46]

$$\dot{u} = -\frac{4 f_{electron} \sigma_T E_{CMB}}{3 m_{proton} c} u = -\gamma u \tag{88}$$

where γ is the viscosity parameter due to Compton dragging, E_{CMB} being the energy density of the CMB photons, $f_{electron}$ is the fraction of the electrons and u is the proper velocity of the fluid. Any inhomogeneity in the CMB temperature is ignored. The three major equations governing the Bondi accretion viz. conservation of mass, conservation of momentum and equation of state of the gas respectively, are

$$\begin{aligned} \frac{dM}{dt} &= 4\pi R^2 u\rho \\ u\frac{du}{dR} + \frac{1}{\rho}\frac{dp}{dR} + \frac{GM(<R)}{R^2} + \gamma(z)u &= 0 \\ p = \omega\rho^\beta \qquad ; \qquad 1 \leq \beta &\leq 5/3 \end{aligned} \tag{89}$$

where, ρ, R, $\frac{dM}{dt}$ are the density of the gas, the radial distance from the PBH and accretion rate respectively. Equations for Bondi accretion are valid only for a quasi-steady accretion. This condition translates to the following mathematical expression

$$t_{crossing} << t_H \qquad \Rightarrow \qquad \frac{R_B H}{c_{s,\infty}} << 1 \tag{90}$$

where, $t_{crossing} \approx \frac{R_B}{c_{s,\infty}}$ and t_H are the sound crossing time and Hubble time respectively. As long as the condition (90) is satisfied, the equations of Bondi accretion can be safely employed to study PBH accretion. The moment $\frac{R_B H}{c_{s,\infty}}$ becomes of the order one, the system does not remain quasi-steady and equations of Bondi accretion no longer remain applicable.

As already mentioned, the "viscous force" gets contribution from both Compton scattering and Hubble expansion. So far, we have only incorporated Compton drag into the Bondi accretion equations via viscosity term $\gamma(z)$. To include Hubble friction as well, we simply express the equations (89) in terms of comoving radius, X, and peculiar velocity u_p

$$
\begin{aligned}
X &= \frac{R}{a(t)} \\
u &= \frac{d}{dt}(Xa(t)) = HR + u_p
\end{aligned} \tag{91}
$$

Substituting (91) into (89) and ignoring the self-gravity of the already accreted gas we get

$$
\begin{aligned}
\frac{dM}{dt} &= 4\pi R^2 u_s \rho(z) \\
u_s\frac{du_s}{dR} + \frac{1}{\rho}\frac{dp}{dR} + \frac{GM(<R)}{R^2} + \gamma_{eff}(z)u_s &= 0 \\
p = \omega\rho^{\beta} \qquad ; \qquad 1 &\leq \beta \leq 5/3
\end{aligned} \tag{92}
$$

Notice that the density ρ_∞ and sound speed $c_{s,\infty}$ at infinity have become a function of redshift and γ is replaced by $\gamma_{eff} = \gamma + H$. This implies that the form of the equations remains the same except for an additional term that gets added to the viscosity parameter.

Unlike baryons, the assumption of neglecting self accretion due to dark matter is not possible if $f_{PBH} < 1$. This is because PBH could accrete as much dark matter as 100 times their initial masses in the matter-dominated era, as already seen earlier. We now focus on the accretion onto a PBH first for a point mass object and then for an extended dark matter surrounding the point mass.

Bondi accretion for a point mass PBH

For the sake of convenience, the accretion equations (92) are rewritten in terms of the dimensionless variables $X = \frac{R}{R_B}$, $\rho_1 = \frac{\rho}{\rho_\infty}$, $c_{s1} = \frac{c_s}{c_{s,\infty}}$ and $U_s = \frac{u_s}{c_s}$ such that

$$\lambda = \rho_1^{\frac{\beta+1}{2}} U_s X^2$$
$$\frac{\dot{U}_s}{U_s} = \frac{\frac{2}{x}((\frac{\beta-1}{2})U_s^2 + 1) - \frac{\beta+1}{2}\left(\frac{1}{c_{s1}^2 X^2} + \frac{\gamma_1 U_s}{c_{s1}}\right)}{U_s^2 - 1} \tag{93}$$
$$c_{s1} = \rho_1^{\frac{\beta-1}{2}}$$

where,

$$\dot{M} = \frac{dM}{dt} = \lambda(4\pi R_B^2 \rho_\infty c_{s,\infty})$$
$$R_B = \frac{GM}{c_{s,\infty}^2}, \tag{94}$$
$$\gamma_1 = \gamma_{eff}\frac{R_B}{c_{s,\infty}}$$

Consider the PBH to be present inside a uniform fluid filled with hydrogen gas and moves with a relative velocity v_{rel} with respect to the surrounding uniform fluid. In such a scenario, the Bondi accretion equation takes the following form

$$\dot{M}_B = \lambda(4\pi R_B^2 m_h n_g u_{eff}) \tag{95}$$

where, n_g, r_B and u_{eff} are the mean cosmic number density, Bondi radius (similar to the second Equation in (94)) and effective velocity respectively, described as

$$n_g \approx 200cm^{-3}\left(\frac{1+z}{1000}\right),$$
$$R_B = \frac{GM}{v_{eff}^2} \approx 1.3\times 10^{-4}pc\left(\frac{M}{M_\odot}\right)\left(\frac{v_{eff}}{5.7km/s}\right)^{-2}, \tag{96}$$
$$u_{eff}^2 = u_{rel}^2 + c_s^2.$$

Bondi radius is physically a very important quantity that defines an apparent spatial boundary beyond which the material does not get accreted onto the central object. This is because the gravitational pull cannot dominate the radiation pressure outside the Bondi radius. Accretion eigenvalue λ is one of the most important quantities for analyzing the accretion phenomenon. It includes all the "opposing terms" that reduce the accretion onto the PBHs namely Compton dragging due to CMB-gas coupling, expanding FRW background and the viscosity of the accreting gas. Reference [42] has calculated the analytical formula for λ

$$\lambda = exp\left(\frac{9}{6+2\gamma_1^{0.75}}\right)X_{cr}^2 \tag{97}$$

where, x_{cr} is the sonic radius given by

$$X_{cr} = \frac{R_{cr}}{R_B} = \frac{(1+\gamma_1)^0.5}{\gamma_1} \tag{98}$$

As already mentioned, the gas viscosity parameter gets contribution from Hubble expansion as well as Compton scattering however, if $\frac{R_B H}{c_{s,\infty}} << 1$, we can ignore the contribution of cosmic expansion to the viscosity parameter. The analytical expression for γ_1 is given by [43]

$$\gamma_1 = \left[0.257 + 1.45 \left(\frac{x_e}{0.01}\right) \left(\frac{z+1}{1000}\right)^{5/2}\right] \times \left(\frac{M}{10^4 M_\odot}\right) \left(\frac{z+1}{1000}\right)^{3/2} \left(\frac{c_s}{5.74 km/s}\right)^{-3} \tag{99}$$

Bondi accretion onto an extended Dark Matter halo

From the current observations, we know that PBHs are unlikely to make up the entire dark matter energy density ($f_{PBH} < 1$) for the majority of the mass range [25]. This implies that there has to be a sizeable amount of independent dark matter fluid causing it to form a halo around the PBHs. The dark matter fluid in the vicinity of the PBHs gets influenced by the gravitational potential of the PBHs and starts getting accumulated. For the sake of simplicity, the halo is assumed to be following a power-law density profile $\rho \propto \frac{1}{R^\alpha}$, where α has an approximate value of 2.25 [47]. Suppose the mass of the accumulated dark matter halo around the PBH is M_h and extended up to the radial distance R_h from the PBH. Analogous to the analytical results (87), the numerical calculations provide the following expression for the growth of halo mass as a function of redshift [41]

$$\frac{M_h}{M} \approx 3 \left(\frac{1000}{1+z}\right) \tag{100}$$

at the same time, its size grows as [47]

$$R_h = 0.019 pc \left(\frac{1000}{1+z}\right) \left(\frac{M}{M_\odot}\right)^{1/3}. \tag{101}$$

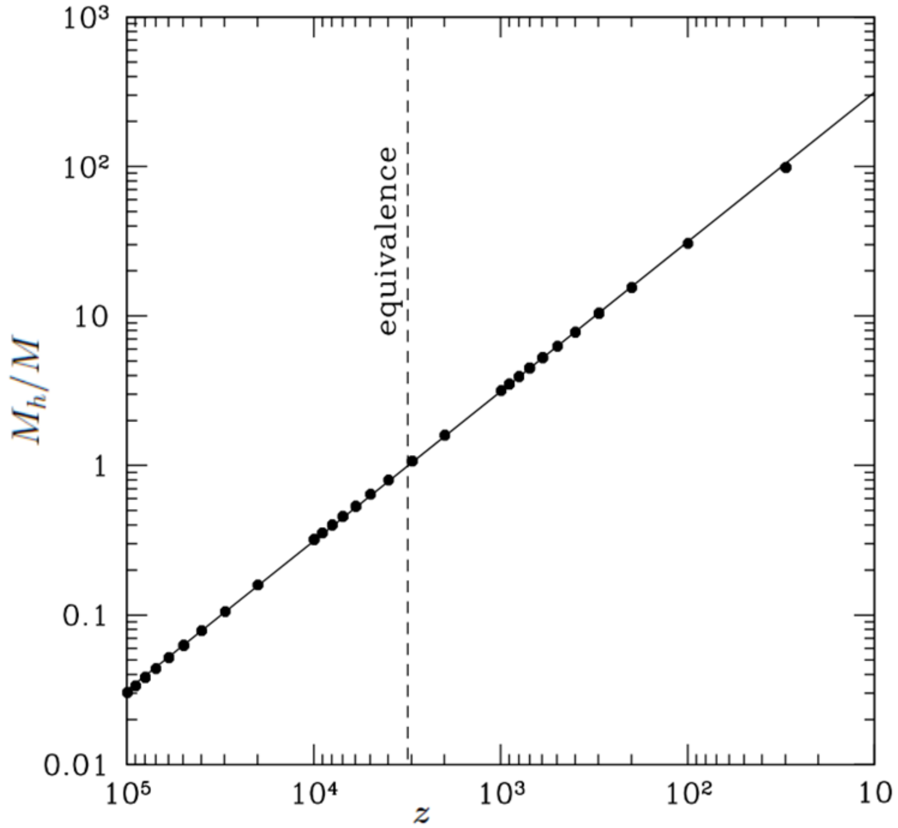

Figure 8: Above schematic plots the mass being accreted as a function of the redshift (z). Vertical dashed line denotes the period of matter-radiation equality. Image credits [41].

Note that these relations have been derived by considering the PBHs as isolated point masses. The accretion stops when almost entire dark matter has been consumed by the halo system. This happens when the redshift of the universe approximately reaches

$$z \approx 1000(3f_{PBH} - 1) \tag{102}$$

Depending upon the extent of the halo radius R_h, the Bondi accretion onto the PBH surrounded by the dark matter halo can be categorized into two schemes. If the halo radius is very less than the Bondi radius, the PBH+dark matter halo can be approximated as a single point mass system and we can study the accretion onto such system with the help of Equation (95), however, the PBH mass has to be replaced by the halo mass $M = M_h$. On the other hand, if the halo radius exceeds the Bondi radius, one must introduce some corrections to the formula (95). It is obvious that the relative fraction of the halo radius and Bondi radius will be the deciding factor for these corrections. Hence, a new parameter

$$\kappa = \frac{R_B}{R_h} = 0.22 \left(\frac{M_h}{M_\odot}\right)^{2/3} \left(\frac{1+z}{1000}\right) \left(\frac{u_{eff}}{km/s}\right)^{-2} \tag{103}$$

is defined. For $\kappa \geq 2$, we can safely use Equation (95) after replacing the PBH mass with the halo mass and the accretion eigenvalue is analogous to Equation (97). For the other case i.e. $\kappa < 2$, the following corrections are introduced in the halo viscosity, halo accretion eigenvalue and halo critical radius parameters respectively,

$$\begin{aligned} \frac{\gamma_1^h}{\gamma_1} &= \kappa^{\frac{p}{1-p}}, \\ \frac{\lambda_h}{\lambda(\gamma_1)} &= \Gamma^{\frac{p}{1-p}}, \\ \frac{R_{cr}^h}{R_{cr}} &= \left(\frac{\kappa}{2}\right)^{\frac{p}{1-p}}, \end{aligned} \tag{104}$$

where,

$$\begin{aligned} p + \alpha &= 2, \\ \Gamma &= (1 + 10\gamma_1^h)^{1/10} \exp(2-\kappa) \left(\frac{\kappa}{2}\right)^2 \end{aligned} \tag{105}$$

7 Formation of the PBH binaries

With the advent of ultra-sensitive instruments of the LIGO and VIRGO interferometers, it has now become possible to observe the mergers of the black hole binaries. By assuming that the binaries are of primordial origin, it is also possible to establish the constraints on their merger rate, abundance and their present dark matter fraction. In this section, we review two different formation mechanisms of PBH binaries that are classified on the basis of the cosmological time of their formation.

7.1 Binary formation in the Early universe

For the early type PBH binaries, their formation happens roughly during the matter radiation equality. Nakamura et al. had given the earliest and simplest formalism to understand the PBH binary formation and even calculated their merger distribution function [49]. This work had assumed that the PBH population constituted the entire fraction of the dark matter density i.e. $f_{PBH} = 1$, the generated PBHs have a monochromatic mass function accompanied by a uniform spatial distribution and a neighbouring PBH (also called a perturber PBH) that generates a necessary tidal force to prevent the head-on collision of the binaries. This analysis has been recently generalized in Reference [55] for the case of an arbitrary value of f_{PBH} and tidal effects due to matter fluctuations during the inflationary epoch. We follow a simplified approach given in Reference [50, 52].

The Decoupling Condition

Let us first derive the criteria for the decoupling of the PBHs from the background. Assume that a pair of PBHs is expanding along with the Hubble background that also contains several other PBHs distributed uniformly. The Hubble parameter is given by

$$H^2 = \frac{8\pi\rho_{tot}}{3} \tag{106}$$

In the early universe, only matter and radiation energy densities are important hence, $\rho_{tot} = \rho_m a^{-3} + \rho_r a^{-4}$ in terms of the comoving densities. In terms of the comoving Newtonian potential ϕ, the line element of the spacetime containing perturbations up to first order can be written as [56]

$$ds^2 = -(1 + 2\phi(x))dt^2 + a(t)^2(1 - 2\phi(x))dx^2, \tag{107}$$

where Einstein equations can be used to obtain the expression for the potential $\phi(x)$ such that

$$\frac{1}{a(t)^2}\Delta\phi = 4\pi\rho_{PBH}(x) \tag{108}$$

where ρ_{PBH} is the energy density attributed to the PBHs. Equation (108) shows that it is a function of position because the masses of PBHs are concentrated only at the points rather than being distributed uniformly. Therefore, Equation (108) can be integrated to give

$$\phi(x) = \sum_i \frac{-M_i}{a(t)|x_i - x|} \tag{109}$$

where x_i and M_i are the comoving position and mass of the i^{th} PBH respectively. To understand the behaviour of the null or timelike geodesics around any system or mass, it is necessary to find out its action S. The action of N PBHs (in the non-relativistic limit) is given a summation of different terms as follows

$$S_N = \int \left\{ \sum_{i>j} \frac{M_i}{|r_i - r_j|} + \sum_i \left(\frac{M_i \dot{r}^2}{2} + \frac{M_i \ddot{a}}{2a} r_i^2 - M_i \phi_0(x) \right) \right\} dt, \qquad r = a(t)x \tag{110}$$

here, the first term in the action denotes a pairwise interaction of the i^{th} and j^{th} PBHs, second and third terms account for the kinetic energy of the system due to the intrinsic PBH velocity and Hubble expansion respectively. The last potential term shows the influence of the surroundings on the system. We are concerned with a system of binaries hence, for $N = 2$ the Equation (110) becomes

$$S_2 \approx \int \left\{ \frac{\mu}{2} \left(\dot{r}^2 + \frac{\ddot{a} r^2}{a} + \frac{2M_{tot}}{\mathbf{r}} - r \cdot \tau \cdot r \right) + \frac{M_{tot}}{2} \left(\dot{r_c^2} + \frac{\ddot{a}}{a} r_c^2 \right) - M_{tot} \phi_0(r_c) \right\} dt \tag{111}$$

where, $\mu = M_1 M_2 / M_1 M_2$, $M_{tot} = M_1 + M_2$, r and r_c are defined as the reduced mass and total mass of the two PBH system, physical radial separation between PBHs and centre of mass coordinate if the two PBH system respectively. The term τ incorporates all the tidal forces due to the matter surrounding the pair. It is convenient to write down all the forces acting on the two-PBH system to evaluate the condition for decoupling

$$\mathcal{F} = \left(-\frac{M_1 M_2}{r^2} + \mu \frac{\ddot{a}}{a} r \right) \hat{r} + f(\tau)(\hat{r}, \hat{\theta}) \tag{112}$$

The third term is an explicit function of the tidal force in both radial and angular directions. The first and second terms denote the gravitational pull and the "Hubble friction" force respectively. The tidal forces have to be much weaker than the gravity term otherwise the PBH pair would get disrupted rather than forming a binary. In addition, if the system is studied in the rest frame of its centre of mass then the time dependence of the tidal forces arises only due to the expanding background.

The gravitational force pulls the PBH pair closer while the expansion drives them away. The PBH pair is said to have decoupled from the Hubble flow when the first term becomes much larger than the second term i.e.

$$\frac{M_1 M_2}{r^2} >> \mu \frac{\ddot{a}}{a} r \tag{113}$$

rearranging the terms and replacing physical radius with the comoving radius,

$$\frac{M_{tot}}{x^3} >> \left|\frac{\ddot{a}}{a}\right| a^3, \tag{114}$$

second Friedmann equation gives

$$\frac{M_{tot}}{2V(x)} >> \left|-\rho_m - 2\frac{\rho_r}{a}\right| \Rightarrow \frac{M_{tot}}{2\rho_m V(x)} >> \left|-1 - 2\frac{\rho_r}{\rho_m a}\right| \tag{115}$$

which eventually leads to the necessary condition for a system of two PBHs to decouple from expansion

$$\Delta_B = \frac{M_{tot}}{2\rho_m V(x)} >> 1 \tag{116}$$

Distribution Function

The average physical separation of the PBHs is given by

$$X_{avg} = \left(\frac{M}{\rho_{BH}(z_{eq}}\right)^{1/3} = \left(\frac{8\pi GM}{3(1+z_{eq})^3 f_{PBH} H_0^2 \Omega_{DM}}\right)^{1/3} \tag{117}$$

where the energy density of the PBHs ρ_{PBH} is calculated at the matter-radiation equality (z_{eq}). Consider a pair of nearby PBHs separated by a physical distance X such that $X < X_{avg}$. Once decoupled, the PBH pair forms a closed binary system thanks to the tidal forces. Thereafter the binary follows its own mechanics and becomes independent of the expansion. The decoupling happens at the redshift (z_{dec}) given by

$$z_{dec} + 1 = (1 + z_{eq})(f_{PBH}\left(\frac{X_{avg}}{X}\right)^3 - 1) > 0 \tag{118}$$

To understand the dynamics of a PBH binary, the first step should be to define the orbital parameters for the binary. Suppose Y is the physical distance of the nearest perturber PBH from the centre of mass of the binary at the time of its formation then the initial magnitudes of the minor axis b_i and major axis a_i are given by [52]

$$b_i = \left(\frac{X\beta^3}{Y}\right)^3 a_i, \qquad a_i = \frac{\alpha X^4}{f_{PBH} X_{avg}^3} \tag{119}$$

where, the values of the coefficients α and β have to be determined with the help of numerical simulations. The initial eccentricity e_i of the binary can then be calculated such that

$$e_i = \sqrt{1 - \frac{b_i^2}{a_i^2}} = \sqrt{1 - \frac{\beta^{12} X^6}{Y^6}} \tag{120}$$

We follow the Reference [52] and take $\alpha = \beta = 1$ although the results do not create much difference if one considers the numerically calculated values $\alpha = 0.4$ and $\beta = 0.8$ [57]. Note that the distance between the perturber PBH and the binary's centre of mass has to be less than the average distance average physical separation between the PBHs. At the same time, PBHs forming binary must be closer to each other than the perturber PBH. Expressing these two conditions mathematically we get

$$Y < X_{avg} \qquad \& \qquad Y > X \tag{121}$$

The first condition in equation (121) gives the maximum allowed value of the eccentricity

$$e_{i,max} = \sqrt{1 - f_{PBH}^{3/2} \left(\frac{a_i}{X_{avg}}\right)^{3/2}} \tag{122}$$

Assuming a uniform spatial 3 dimensional PDF distribution for Y and X the differential probability (dP) of finding a PBH binary whose separation lies between $(X, X + dX)$ and simultaneously finding a perturber PBH whose distance from the centre of mass of the binary lies between $(Y, Y + dY)$ is

$$dP = \frac{9X^2Y^2dXdY}{X_{avg}^6} \tag{123}$$

Note that if the PBHs start clustering on account of their mutual gravitational attraction, the assumption of uniform spatial PDF no longer remains valid. It is desirable to express the differential probability in the Equation (123) as a function of the orbital parameters a_i and e_i which can be done using Equations (119) and (120) to get

$$dP(e_i, a_i) = \frac{3}{4} f_{PBH}^{3/2} a_i^{1/2} X_{avg}^{-3/2} e_i (1 - e_i^2)^{-3/2} de_i da_i \tag{124}$$

Equation (124) can be easily integrated to obtain the differential probability as a function of a single variable. It is clear that the probability of finding binaries with large eccentricities is higher.

Merger Rate

Once the binary is formed, it loses energy through the emission of gravitational waves. Hence, the orbit shrinks which eventually leads to the merger of the binary. If the effects of gravitational wave emission are ignored, the orbital parameters a_i and e_i can be approximated as the constants of motion. Following Reference [54] we calculate the merger rate for the PBH binaries formed in the early universe. As already mentioned, more eccentric binaries have a higher probability of formation. On top of that, the external tidal torque due to surrounding matter is much weaker as compared to the mutual gravitational pull of the binaries and hence, the eccentricity of the orbit tends to be much closer to unity. For a PBH binary consisting of masses M_1 and M_2, the coalescence time for a nearly elliptic orbit ($e_i = 1$) is given by [60]

$$t_c = \frac{3}{85} \frac{a_i^4 (1 - e_i^2)^{7/2}}{G^3 (M_1 + M_2)^2 \mu} \tag{125}$$

To calculate the merger rate of the binaries, we need to express the differential probability distribution Equation (124) as a function of the coalescence time

(t_c). Therefore, it is required to rewrite the Equation (125) in the form of orbital parameters such that

$$t_c(a_i, e_i) = \frac{3a_i^4}{170G^3 M_{PBH}^3}(1 - e_i^2)^{7/2} \tag{126}$$

We can now eliminate the semi major axis a_i from the Equation (124) as a variable using Equation (126) so that the differential probability distribution is expressed as a function of binary coalescence time and its eccentricity

$$dP = \frac{3e}{16(1-e^2)^{45/16}} \left(\frac{170G^3 M_{PBH}^3 f_{PBH}^4 t_c}{3X_{avg}^4} \right)^{3/8} \frac{dt_c}{t_c} de_i \tag{127}$$

The upper limit of the eccentricity is given by

$$\begin{aligned} e_{i,up} &= \sqrt{1 - \left(\frac{170G^3 M_{PBH}^3 f_{PBH}^4 t_c}{3X_{avg}^4} \right)^{6/37}} \quad \text{if} \quad t < T \\ e_{i,up} &= \sqrt{1 - f_{PBH}^2 \left(\frac{t}{T} \right)^{2/7}} \quad \text{if} \quad t \geq T \end{aligned} \tag{128}$$

where,

$$T = f_{PBH}^{25/3} \frac{3X_{avg}^4}{170G^3 M_{PBH}^3} \tag{129}$$

Integrating Equation (128) with respect to e_i we get the differential probability of the binary having the coalescence time between $(t_c, t_c + dt_c)$

$$dP_{t_c} = \frac{3}{58}\left\{\left(\frac{170G^3M_{PBH}^3f_{PBH}^4t_c}{3X_{avg}^4}\right)^{3/37} - \left(\frac{170G^3M_{PBH}^3f_{PBH}^4t_c}{3X_{avg}^4}\right)^{3/8}\right\}\frac{dt_c}{t_c} \quad \text{if} \quad t < T$$
$$dP_{t_c} = \frac{3}{58}\left(\frac{170G^3M_{PBH}^3f_{PBH}^4t_c}{3X_{avg}^4}\right)^{3/8}\left\{\left(\frac{t_c}{T}\right)^{-29/56} - 1\right\}\frac{dt_c}{t_c} \quad \text{if} \quad t \geq T \tag{130}$$

Finally, the merger rate per unit time per unit volume is given by

$$\mathfrak{R}_{PBH} = n_{PBH}\frac{dP_{t_c}}{dt_c} \tag{131}$$

where n_{PBH} is the mean number density of the PBH today and given by

$$n_{PBH} = \frac{3\Omega_{PBH}H_0^2}{8M_{PBH}\pi G} \tag{132}$$

The reason why the merger rate is so important is that it is the quantity that can actually be constrained with the help of LIGO-VIRGO detectors and thus, the merger rate is one of the best tools to verify the existence and abundance of the PBHs. The present estimate on the merger rate at redshift $z = 0.09$ from the LIGO-VIRGO collaboration is $2 - 53Gpc^{-3}yr^{-1}$ [61].

7.2 Binary formation in the late universe

In the late universe, the PBHs have high velocities due to the high gravitational potential of the already formed halos. On top of that, they are coupled to the Hubble flow which makes it unlikely for two PBHs to form a binary. Although, it is theoretically possible for two PBHs to come under the influence of one another provided they pass sufficiently close to each other. Following the approach of References [51, 54] we summarize the formation of PBH binaries in the late universe.

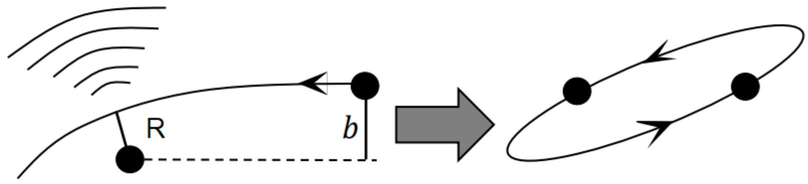

Figure 9: Picture shows two PBHs approaching each other with an impact parameter b. Image credits [54]

As shown in the Figure 9, consider a system of two PBHs, one of which is stationary and the other is moving with a velocity V, having a close encounter with periastron (R) and impact parameter (b). As the masses of these PBHs are concentrated at a single point, the probability of their head-on collision is very less. Therefore, they are likely to achieve a little angular momentum. When the PBHs reach the periastron of the trajectory their relative acceleration reaches its maximum value and results in the energy loss of the system through the emission of gravitational waves. The magnitude of the rate of time-averaged energy loss via gravitational waves has been calculated in [60]

$$\left\langle \frac{dE}{dt} \right\rangle = -\frac{32}{5} \frac{G^4 M_1^2 M_2^2 (M_1 + M_2)}{a_i^5 \left(1 - e_i^2\right)^{7/2}} \left(1 + \frac{73}{24} e_i^2 + \frac{37}{96} e_i^4\right) \tag{133}$$

The orbit is assumed to be Keplerian. Assuming the masses of the PBHs to be equal, this equation can be integrated over one time period of the orbit to get calculate the energy loss due to GWs after one revolution such that

$$\Delta E = \frac{85\pi}{12} \frac{G^{7/2} M^{9/2}}{R^{7/2}} \tag{134}$$

The impact parameter in the Newtonian approximation turns out to be [59]

$$b(R) = \sqrt{R^2 + \frac{2GMR}{V}} \tag{135}$$

A necessary condition for the formation of the binary is that the kinetic energy of the binary system has to be less than the energy emitted through the gravitational waves. This condition puts a constraint on the magnitude of the periastron which is given by

$$R < R_{max} = \frac{85\pi}{3} \frac{G^{7/2} M^{7/2}}{V^2} \tag{136}$$

Thus, the maximum value of the impact parameter for a binary to form is $b(R_{max})$. For the case $R << b$, also known as strong gravitational focusing, it is possible to approximate the cross-section (σ_{PBH}) for the formation of a PBH binary

$$\sigma_{PBH} \simeq \left(\frac{85}{3}\right)^{2/7} \frac{\pi^{9/7} (2GM)^2}{V^{18/7}} \tag{137}$$

Merger Rate

The earliest calculations for the merger rate PBH binaries formed in the late universe, when the halos have already, been discussed in [58]. However, the velocities of PBHs are significantly large in the late universe and according to Equation (137), the cross-section for the binary formation of such PBHs would be very low. Nonetheless, it is theoretically possible for the PBHs to form a binary if they pass considerably close to each other and when they do, the emission of gravitational waves takes place owing to their interaction that results in the formation of a closed system. It is also obvious from the Equation (137) that binary formation is more efficient for the slow-moving PBHs and in the high-density regions where the probability of PBHs passing close to each other is more. This implies that the formation of the PBH binaries finds a suitable environment inside halos having low mass. The rate of merger for PBH binaries formed within a dark halo with mass M_H can then be calculated as [54]

$$\mathfrak{R}_{H,PBH}(M_H) = 2\pi \int_0^{R_{vir}} \left(\frac{\rho_{PBH}(R)}{M_{PBH}} \right)^2 \langle \sigma_{PBH} V \rangle \, R^2 dR \tag{138}$$

where the density of the PBHs inside the halo, $\rho_{PBH}(R)$ has been assumed to be following the Navarro-Frenk-White spatial distribution [58]. The term inside brackets is the product of the binary cross-section and relative velocity is averaged over the velocities obeying the Maxwell-Boltzmann distribution function. It is now possible to compute the total merger rate of the PBH binaries

$$\mathfrak{R}_{PBH} = \int_{M_{ex}} \frac{dn}{dM_H} \mathfrak{R}_{H,PBH} dM_H \tag{139}$$

The lower limit on the integral M_{ex} is the minimum value of the PBH mass that could sustain the Hawking evaporation till the present time. The halo mass function $\frac{dn}{dM_H}$ that gives information about the distribution of the halo as a function of its mass, has been computed using the Press-Schechter formalism and given by

$$\mathfrak{R}_{PBH} \approx 2 f_{PBH}^{53/21} Gpc^{-3} yr^{-1} \tag{140}$$

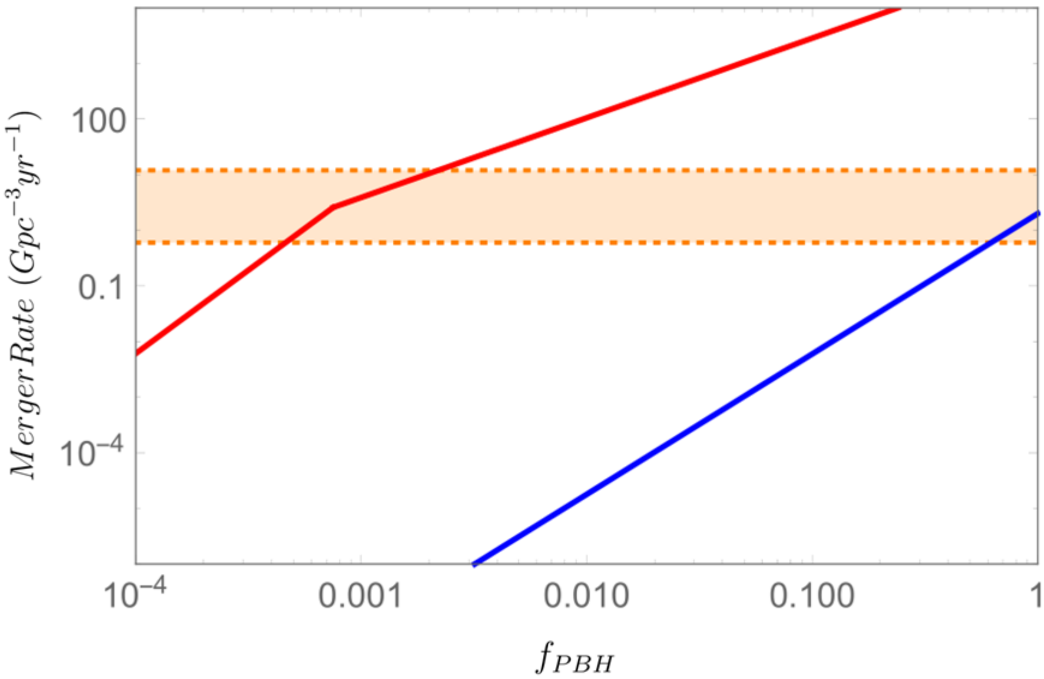

Figure 10: The plot shows the total merger rate of the PBH binaries as a function of their mass fraction (f_{PBH}). Red curve [52] and Blue curve [58] show the merger rates of the binaries formed during early times and late times respectively. The orange band represents the present bounds on the merger rate from LIGO-VIRGO collaborations [62].

It is evident from Figure 10 that the merger rate of PBH binaries formed in the late universe falls in the LIGO-VIRGO window only when PBHs make up the entire portion of the dark matter ($f_{PBH} \approx 1$). Notice that as compared to the merger rate of early time PBH binaries, the merger rate of late time PBH binaries is very low and it is even lower for $f_{PBH} << 1$. Interestingly, Reference [64] has pointed out that if PBHs clustering becomes prominent near the supermassive black holes. It is possible that late time PBHs binaries may become the dominant source of the gravitational waves as their total merger rate could become more than (140) for a specific type of density profile of the PBHs that are in the vicinity of the supermassive black holes.

8 Accretion Onto Binary PBHs

Once the PBHs form binaries, the accretion mechanism has to be studied by considering them as a two-body system. The binary accretion is categorized into two processes viz. a local accretion process, where the accretion occurs onto the binary PBHs individually, and, a global accretion process, which happens when the binary accretes as a single system.

As shown in the Figure 11, assume a binary consisting of two PBHs having unequal masses M_1 and M_2 such that $M_1 > M_2$. The reduced mass of the binary, its total mass and binary mass ratio are defined as $\mu = M_1 M_2/M_1 + M_2$, $M_{tot} = M_1 + M_2$ and $q = M_2/M_1$ respectively. The binary mass ratio is always less than or equal to unity.

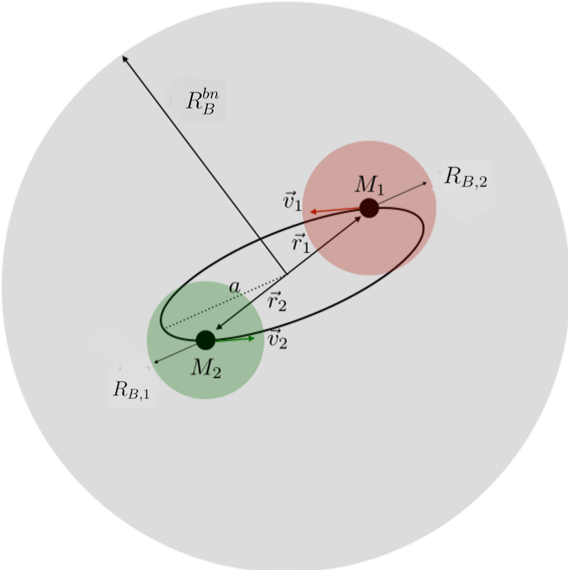

Figure 11: The figure shows a binary consisting of two PBHs having unequal masses and thus, unequal Bondi radii. The individual Bondi radii of the PBHs are shown with green and red colours whereas the grey area depicts the Bondi radius of the binary as a whole. The semi-major axis and the Bondi radii follow the inequality $M_i << R_B \lessapprox a_i << R_B^{bn}$. [101]

This classification of binary accretion (global and local) is based on the relative size of the overall Bondi radius of the binary R_B^{bn}, which is expressed as a function of the effective velocity v_{eff} such that

$$R_B^{bn} = \frac{M_{tot}}{v_{eff}^2}; \qquad v_{eff} = \sqrt{v_{rel}^2 + c_s^2} \tag{141}$$

The relative size ratio of the semi-major axis a and the Bondi radius R_B^{bn} of the binary is the factor that determines which kind of accretion will dominate:

⇒ If $a >> R_B^{bn} 1$, the distance between PBHs in the binary is very large and their individual Bondi radii are too far beyond each other's reach. Therefore, both the PBHs accrete separately without interfering with the accretion process of one another. At the same time, the orbital motion affects the characteristic velocities of the PBHs relative to the background fluid.

⇒ If $a << R_B^{bn}$, the binaries are so close to each other that the accreting fluid coming from a large distance cannot differentiate the two closely packed PBH binaries. Hence, the system accretes as if it were a single PBH of mass M_{tot} following the standard Bondi accretion process which can be described by

$$\dot{M}_{bn} = 4\pi m_H n_{gas} v_{eff}^{-3} M_{tot}^2 \tag{142}$$

The first case is not much relevant to us because the binaries would be too far away from each other to merge within the age of the universe. Hence, we focus on the second case. Coming to the orbital parameters, the characteristic velocities and instantaneous of the PBHs with respect to their centre of mass have been calculated to be [66]

$$v_1 = \frac{qv}{1+q}, \quad v_2 = \frac{v}{1+q}, \quad r_1 = \frac{qr}{1+q}, \quad r_2 = \frac{r}{1+q} \tag{143}$$

where the velocity v and their relative separation r is given by

$$v = \sqrt{M_{tot}\left(\frac{2}{r} - \frac{1}{a}\right)}, \qquad r = a(1 - e\cos u) \tag{144}$$

here u is the orbital angle that changes with time and its expression is given by

$$u(t) = \sqrt{\frac{M_{tot}}{a^3}}(t - T) + e \sin u(t) \tag{145}$$

where T is the constant of integration. The effective velocities of the PBHs relative to the background fluid are expressed as

$$v_{eff,1}^2 = v_1^2 + v_{eff}^2, \qquad v_{eff,2}^2 = v_2^2 + v_{eff}^2 \tag{146}$$

As already mentioned, we are only concerned with the case where $a << r_B^{bn}$. The semi major axis being much smaller than the binary's Bondi radius, the rate of infall of the accreting baryons can be considered constant or

$$\dot{M}_{bn} = 4\pi n_{gas}(R) m_H v_{ff}(R) R^2 = constant \tag{147}$$

where $n_{gas}(R)$ is the fluid density profile as a function of the radial distance R from the binary's centre of mass

$$n_{gas}(R) = \dot{M}_{bn}/4\pi v_{ff}(R) m_H R^2 \tag{148}$$

It is evident from the Equation (148) that as the accretion rate onto the binary increases, the surrounding environment starts getting denser and denser. At the same time, it decreases as we move away from the centre of mass of the binary. The freefall velocity of the infalling fluid, v_{ff} is given by

$$v_{ff}(R) = \sqrt{\frac{2M_{tot}}{R} + v_{eff}^2 - \frac{2M_{tot}}{R_B^{bn}}} \tag{149}$$

which is defined by assuming that it reduces to the effective velocity v_{eff} at large distances. For the case of individual accreting PBHs, the respective accretion rates are given as

$$\begin{aligned}\dot{M}_1 &= 4\pi n_{gas}(R_{B,1}) m_H v_{eff,1}^{-3} M_1^2 \\ \dot{M}_2 &= 4\pi n_{gas}(R_{B,2}) m_H v_{eff,2}^{-3} M_2^2\end{aligned} \tag{150}$$

The accretion parameter λ is approximated to be one at low redshifts. On the other hand, for the case of accretion as a whole, the λ parameter incorporates the ratio of the binary Bondi radius and the halo radius of the surrounding Dark Matter i.e. R_B^{bn}/R_H. As explained in the previous section, this ratio determines whether the accreting object+halo needs a correction in their equations or not. In the case of PBHs binaries accreting individually, they cannot sustain the surrounding halo and hence accrete as naked PBHs, hence justifying the choice of $\lambda \approx 1$. It is desirable to express the accretion formulas (150) as a function of the orbital parameters. After some tedious calculations one arrives at [53],

$$\begin{aligned}\dot{M}_1 &= \dot{M}_{bn}\sqrt{\frac{1+\chi_1+(1-\chi_1)\chi_2^2}{2(1+\chi_1)(1+q)+(1-\chi_1)(1+2q)\chi_2^2}} \\ \dot{M}_2 &= \dot{M}_{bn}\sqrt{\frac{(1+\chi_1)q+(1-\chi_1)q^3\chi_2^2}{2(1+\chi_1)(1+q)+(1-\chi_1)(2+q)q^2\chi_2^2}}\end{aligned} \tag{151}$$

Here, the modified orbital parameters χ_1 and χ_2 are defined as

$$\begin{aligned}\chi_1 &= e\cos u \\ \chi_2 &= \sqrt{\frac{a}{\mu q}}v_{eff}\end{aligned} \tag{152}$$

It is possible to make the Equations (152) explicitly time-independent by averaging them over the orbital angle u. Note that we can assume that the change in mass due to accretion is negligible during this averaging because the orbital period is much shorter than the characteristic timescale of accretion. Equations (152) after averaging take the following form

$$\begin{aligned}\dot{M}_1 &= \dot{M}_{bn}\sqrt{\frac{a(1+q)v_{eff}^2 + M_1 q^2}{(1+q)[a(1+2q)v_{eff}^2 + 2M_1 q^2]}} \\ \dot{M}_2 &= \dot{M}_{bn}\sqrt{\frac{q[a(1+q)v_{eff}^2 + M_1]}{(1+q)[a(2+q)v_{eff}^2 + 2M_1]}}\end{aligned} \tag{153}$$

One can write an simpler form of the Equations (153) for the case $av_{eff}^2 << M_1 q^2$. However, this condition is valid only for $a \sim \mathcal{O}(10^6)M_1,\ M_1 \sim \mathcal{O}(M_\odot)$ and $q > 10^{-2}$. The expressions in (153) now become simpler

$$\begin{aligned}\dot{M}_1 &= \dot{M}_{bn}\frac{1}{\sqrt{2(1+q)}}, \\ \dot{M}_2 &= \dot{M}_{bn}\sqrt{\frac{q}{2(1+q)}}\end{aligned} \tag{154}$$

When the binary consists of PBHs having equal masses i.e. for $q \approx 1$, we get $\dot{M}_1 = \dot{M}_2 = \dot{M}_{bn}/2$. An interesting thing about the accretion Equations (154) is that they do not depend on the magnitude of the orbital parameters. Moreover, by just knowing the initial mass ratio of the binary q and the binary accretion rate $\dot{M}_{bn}$, we can get complete information about the evolution of the masses of the PBHs with respect to time. Conventionally, the accretion rate Equations (154) are represented in the form of Salpeter time $T_{Sal} = 4.5 \times 10^8 yrs$ and Eddington rate $\dot{m} = T_{Sal}\dot{M}_j/M_j$ which is given by

$$\begin{aligned}\dot{m}_1 &= \ddot{m}_{bn}\sqrt{\frac{1+q}{2}}, \\ \dot{m}_2 &= \ddot{m}_{bn}\sqrt{\frac{1+q}{2q}}\end{aligned} \tag{155}$$

A little refurbishing of the Equations (154) gives the formula for time evolution of the mass ratio q

$$\frac{\dot{q}}{q} = \left(\frac{\dot{M}_2}{M_2} - \frac{\dot{M}_1}{M_1} \right) = \frac{\dot{m}_2 - \dot{m}_1}{T_{sal}} \tag{156}$$

As long as the small mass PBH M_2 accretes faster than the large mass PBH M_1, the growth rate of the mass ratio is positive with respect to time. It becomes zero (q stops growing) when q becomes one i.e. both the PBHs achieve equal masses. This implies that this kind of accretion model tries to equalize the masses of the PBH binaries.

9 Constraints on the PBH population

In this section, we describe several observational probes that collectively help us in establishing the constraints on the abundance of the PBHs in the present universe and in turn, their contribution to the Dark Matter density. But before that, let us first get ourselves familiar with the standard PBH terminology and PBH formation calculations. We follow the References [67, 68] for the next subsection.

9.1 Standard PBH calculation

PBHs are usually formed in the early radiation dominated era when the scalar fluctuations re-enter the horizon. Carr and Hawking speculated that the mass of the newly formed PBH (M_{PBH}) is of the order of the horizon mass (M_{Hor}) at that time given by [22]

$$M_{PBH} = \gamma M_{Hor} \tag{157}$$

The efficiency factor γ is of the order ~ 0.4 [9, 14]. Suppose the inflationary era is driven by a single scalar field ϕ that is minimally coupled to gravity. The action of such a system is given by

$$\mathcal{S} = \int \left[\frac{m_P^2 R}{2} - \frac{\partial_\mu \phi \partial^\mu \phi}{2} - V(\phi) \right] \sqrt{-g} d^4x \tag{158}$$

where the Planck mass is $m_P = (8\pi G)^{-1/2}$, R is the scalar curvature and $V(\phi)$ is the potential of the scalar field ϕ, also known as the Inflaton field. As already mentioned, inflation lasts for an extremely short time so, it is not favourable to express the evolution equations of the inflationary phase in terms of cosmic time. Instead, one uses $N(t)$, known as the number of e-folds before the end of inflation, given by

$$\frac{dN}{dt} = H(t) \qquad \Rightarrow \qquad a(t) = e^N \tag{159}$$

The slow roll parameters ϵ_{SR} and η_{SR} must be very less than unity and their explicit expression is given as [2]

$$\begin{aligned} \eta_{SR} &= \frac{-\ddot{\phi}}{H\dot{\phi}} << 1, \\ \epsilon_{SR} &= \frac{-\dot{H}}{H^2} << 1 \end{aligned} \tag{160}$$

It is convenient to write the Klein Gordon Equation (29) and the first Friedmann Equation (13) in terms of the slow roll parameters (160)

$$\begin{aligned} \frac{d^2\phi}{dN^2} + (2-\epsilon)\frac{d\phi}{dN} + \frac{1}{H^2}\frac{dV}{d\phi} &= 0 \\ H^2 &= \frac{1}{(3-\epsilon)m_P^2}V(\phi) \end{aligned} \tag{161}$$

We rewrite the line element for the perturbed spacetime (162) using Φ as the first order scalar perturbation to avoid the confusion with the scalar field ϕ

$$ds^2 = -(1+2\Phi(x))dt^2 + a(t)^2(1-2\Phi(x))dx^2, \tag{162}$$

Note that the vector perturbations decay exponentially with time when the universe is inflating and the tensor perturbations remain nearly constant till the present time. Moreover, both vector and tensor perturbations do not have a source term in the energy-momentum tensor. Hence, they are not important for the study of PBHs and the inflationary phase in general. Coming back to the scalar perturbations, the gauge-invariant form of Φ is more suitable to study the perturbation equations

$$\mathcal{R} = \delta\phi\frac{H}{\dot{\phi}} + \Phi \tag{163}$$

Solving the coupled perturbed Einstein equation (up to first order) solely for the scalar perturbations gives the equation of motion for the gauge-invariant scalar perturbation variable $\mathcal{R}$ given by

$$\frac{d^2\mathcal{R}_k}{dN^2} + (3+\epsilon-2\eta)\frac{d\mathcal{R}_k}{dN} + \frac{k^2\mathcal{R}_k}{e^{2N}H^2} = 0 \tag{164}$$

The most important aspect of the PBH abundance calculation is obtaining the curvature power spectrum $\mathcal{P}_{\mathcal{R}}(k)$ with the help of Equations (161), (164) and by defining the slow-roll parameters as a function of the number of e-folds N. In addition to $\mathcal{P}_{\mathcal{R}}(k)$, we also have a scalar power spectrum for the density fluctuations that are denoted by $\mathcal{P}_S(k)$ and is related to the curvature power spectrum as [69]

$$\mathcal{P}_S(k) = \left(\frac{2k}{3aH}\right)^4 \mathcal{P}_{\mathcal{R}}(k) \tag{165}$$

where the curvature power spectrum as a function of the Fourier modes is given by

$$\mathcal{P}_{\mathcal{R}}(k) = \left[\frac{k^3|\mathcal{P}_{\mathcal{R}}(k)^2|}{2\pi^2}\right]_{k/aH<<1} \tag{166}$$

It is necessary to smoothen the density fluctuations δ on a certain scale $R \sim (aH)^{-1}$ so that the Fourier modes corresponding to small wavelengths are differentiable as well as non-divergent. To achieve this, a window function $W(x, R)$ is used and usually, it is assumed to be a Gaussian PDF. The variance of the density fluctuations in Fourier space after smoothing the density contrast becomes

$$\sigma_S^2(R) = \int_0^\infty \mathcal{P}_S(k)\mathcal{W}^2(k, R)\frac{dk}{k} \tag{167}$$

where $\mathcal{W}^2(k, R)$ is the window function in Fourier space. Coming back to the mass of the PBHs at the time of formation, the horizon mass in Equation (157) is equal to $1/(2GH)$. We are dealing with the radiation dominated epoch so, the Hubble parameter in this epoch is $H = 1/2t$. This gives us the following expression for the PBH mass at the formation

$$M_{PBH} = \frac{\gamma t}{G} \tag{168}$$

The above expression can be obtained as a function of the effective degrees of freedom g_*, efficiency factor γ, wave number at CMB pivot scale k_* and wave number at the time of horizon exit $k_{ex} = 1/R$ such that [67, 68]

$$\frac{M_{PBH}}{M_\odot} = 1.13 \times 10^{15} \left(\frac{\gamma}{0.2}\right) \left(\frac{k_*}{k_{ex}}\right)^2 \left(\frac{g_*}{106.75}\right)^{-1/6} \tag{169}$$

Finally, we introduce the fraction of the universe's mass that collapses into PBHs at the time of the formation of PBHs, which is popularly denoted by $\beta(M_{PBH})$ and defined as

$$\beta(M_{PBH}) = \left[\frac{\rho_{PBH}(M)}{\rho_{total}}\right]_{t=t_i} \tag{170}$$

where the subscript t_i denotes that the mass fraction has been calculated at the time of formation of the PBHs. Another most useful quantity is the present dark matter fraction of the PBHs, $f(M_{PBH})$, given by

$$f(M_{PBH}) = \left[\frac{\rho_{PBH}(M)}{\rho_{DM}}\right]_{t=t_0} \tag{171}$$

In this case, the subscript t_0 means that the fraction is calculated in the present universe. The reason why $f(M_{PBH})$ is so important is that it is a quantity that can be directly constrained from the observations and thus, could give us information about the present population of the PBHs and how much chunk of dark matter could they constitute. It is obvious that the PBH dark matter fraction ($f(M_{PBH})$) is directly dependent on the initial mass fraction of the PBHs ($\beta(M_{PBH})$). Their relation is given by

$$f(M_{PBH}) = 1.68 \times 10^{8} \left(\frac{M_{PBH}}{M_{\odot}}\right)^{-1/2} \beta(M_{PBH}) \tag{172}$$

Next subsection is dedicated to the study of the constraints that different observations have placed on the fraction $f(M_{PBH})$.

9.2 Description Of The Constraints

Finally, we discuss the bounds set up by several observations on the present abundance of the PBHs. We are only concerned with the PBHs whose mass at the formation is $\gtrsim 10^{15} g$ [70]. PBHs whose mass lies below this value would have evaporated well before the present day and hence, they cannot contribute to the present dark matter fraction. Following the references [26, 73] we revisit the primary constraints on $f_{PBH}(M_{PBH})$, assuming a monochromatic mass function.

1. **Evaporation**

   Hawking radiation puts a lower limit on the mass of the PBHs that they need to have in order to survive till the present day i.e. PBHs having masses $M_{PBH} \lesssim M_{*} \sim 10^{15} g$ would have vanished due to the emission of Hawking radiation [74]. For the case $M_{PBH} > 2M_{*}$, it is possible to ignore any change in the PBH mass and compute the photon time-integrated spectrum $\frac{dN^{\theta}}{dE}$ by multiplying the present age of the universe to the instantaneous spectrum which gives [75]

$$\frac{dN^{\theta}}{dE} \propto \begin{cases} E^{3} M_{PBH}^{3} & (E < M_{PBH}^{-1}) \\ E^{2} M_{PBH}^{2} e^{-E M_{PBH}} & (E > M_{PBH}^{-1}) \end{cases}$$

The peak of this spectrum is obtained when $E \sim 1/M_{PBH}$ and its value does not depend on the mass of the PBH. The background photon number density per unit energy can be calculated by integrating the above spectrum over the PBH mass function. For the case of a monochromatic mass function, this results into

$$\eta(E) \propto f(M_{PBH}) \times \begin{cases} E^3 M_{PBH}^2 & (E < M_{PBH}^{-1}) \\ E^2 M_{PBH} e^{-EM_{PBH}} & (E > M_{PBH}^{-1}) \end{cases}$$

The corresponding intensity can be easily calculated using

$$\mathcal{I}(E) = E\eta(E)/4\pi \propto f(M_{PBH}) \times \begin{cases} E^4 M_{PBH}^2 & (E < M_{PBH}^{-1}) \\ E^3 M_{PBH} e^{-EM_{PBH}} & (E > M_{PBH}^{-1}) \end{cases}$$

The intensity also has its peak value $\mathcal{I}_{max}(E) \propto f_{PBH}(M_{PBH})/M^2$ at $E \sim 1/M_{PBH}$. The observed magnitude of the intensity relates with the photon energy E as $I_{obs} \propto E^{-(1+\alpha)}$, with $\alpha \,\epsilon\, [0.1, 0.4]$. Using the inequality relation $\mathcal{I}_{max}(E) < f_{PBH}(M_{PBH}) \times \mathcal{I}_{obs}(E)$ [76]

$$f_{PBH}(M_{PBH}) < 2 \times 10^{-8} \left(\frac{M_{PBH}}{M_*} \right) \tag{173}$$

which is the constraint on present PBH abundance from evaporation. Of course, Equation (173) is valid only for the case of non-evaporating PBHs i.e. for $M_{PBH} > M_*$.

2. **CMB Constraints**

Cosmic Microwave Background or CMB is the earliest relic radiation that was emitted at around $z = 1100$ when the universe cooled down enough so that photons could decouple from the massive particles and roam freely throughout the space. The wavelength of this relic radiation kept on stretching due to the cosmological redshift and the gravitational redshift. Today the electromagnetic spectrum of this radiation is peaked in the microwave region, hence the name Cosmic Microwave Background.

The constraints on the present abundance of PBHs from CMB can be calculated by measuring the magnitude of influence of the radiation emitted from PBH accretion and PBH evaporation on the CMB photons. This radiation emission from PBHs affects the spectrum of CMB in two ways viz.

(a) **Spectral Distortions in CMB**

As CMB is an electromagnetic wave, whose spectral energy distribution peaks in the microwave region, it is expected to follow a Planckian power spectrum. However, due to several cosmological effects in the early universe, the Planckian spectrum of CMB could distort significantly from a perfect black body spectrum. This phenomenon is known as CMB spectral distortion. In 1977, Zeldovich et al. [77] showed that the radiation from PBHs having masses less than $10^9 g$ put very mild constraints on the present PBH abundance. On the other hand, PBHs with masses in the range $10^{13} g > M_{PBH} > 10^{11} g$ have the potential to distort the CMB considerably if their (modified) mass fraction exceeds [25]

$$\beta_{mod}(M_{PBH}) = \left(\frac{M_{PBH}}{m_p}\right)^{-1} \approx \left(\frac{10^{-5}g}{M_{PBH}}\right) \tag{174}$$

(b) **Temperature Anisotropies in CMB Spectrum**

The photons from PBHs can also affect the CMB anisotropy spectrum. Due to the emission of extra energy from PBHs in the early universe, the rate of ionization could increase notably, which leads to the recombination later than the usual time and this causes a shift in the peaks of the anisotropy spectrum [78]. The earliest bounds from the accretion of PBHs [79], which consider the case of spherical accretion, are very weak. It was, however, argued in Reference [80] that spherical accretion is not a realistic case of accretion rather, accretion disks turn out to be much more physical and they put stronger constraints on the PBH abundance than the previous works. These constraints have been further tightened in the Reference [81], where the authors have assumed that the accreting PBHs are surrounded by a dark matter halo that enhances the accretion rate, as discussed in the previous sections. Hence, it is clear that the bounds from accreting PBHs largely depend on the type of accretion that is being considered.

3. **Gravitational Waves Constraints**

The recent detection of multiple black hole mergers by LIGO-VIRGO detectors is the primary reason behind the renewed interest of the cosmology community in the PBHs [61, 82, 83]. In case the black hole mergers turn out to be of stellar origin rather than primordial, the data from these observations will help in establishing the constraints on PBH abundance for a range of masses. Clearly, the efficiency of the constraints from LIGO-VIRGO observations relies on how realistic can we make the mechanism of PBH binary formation and mergers in the radiation dominated era. For PBHs having masses of the order $M_{PBH} \sim 10M_{\odot}$, latest calculations for the merger rates of the PBHs have verified the constraints on the present dark matter fraction of the PBH in the range $f_{PBH} \ \epsilon \ [0.001, 0.01]$ [84]. It should be noted that the authors of Reference [84] have emphasized two important aspects of the PBH binaries viz. the role of initial clustering of the PBHs and the spatial distribution of the 'perturbing' PBHs that provide the necessary tidal force to the binaries. So far, the LIGO-VIRGO detectors have not detected the mergers of black holes for the mass range $M_{PBH} \ \epsilon \ [0.2M_{\odot}, 1M_{\odot}]$ and the PBH dark matter fraction constraint for the corresponding mass range has been calculated to be [85]

$$f < \begin{cases} 0.3 & ; M_{PBH} < 0.2M_{\odot} \\ 0.05 & ; M_{PBH} < 1M_{\odot} \end{cases}$$

Constraints from Gravitational Wave emission on f_{PBH} can also be set up from the density fluctuations that gave rise to the PBHs [86]. The frequency of the corresponding gravitational waves is of the order $\sim 10^{-11} \left(\frac{M_{PBH}}{10^3 M_{\odot}}\right)$ [87]. The observations corresponding to these frequencies have the potential to explore the PBH dark matter fraction for the mass range $\sim 10^{20}g$. Interestingly, this mass range is currently unbounded and carries a possibility that PBHs of approximately similar masses could make up a significant fraction of dark matter [88].

4. **Gravitational Lensing Constraints**

Electromagnetic radiation emitted from a distant object passes through the space that lies between the earth and the object. If there exists a massive foreground object (for example a galaxy) in this intermediate region, the electromagnetic radiation gets deflected from its original path because of the gravitational influence of the foreground object. This phenomenon is referred to as gravitational lensing. Depending on the mass, orientation and dimensions of the foreground object, gravitational lensing can be further divided into various sub-categories viz. strong lensing, weak lensing and microlensing. Among these, strong lensing is generally witnessed when a foreground object is extremely massive, for example, a galaxy or a cluster. and produces multiple images of the source. Weak lensing occurs when the object is not massive enough to create multiple images rather, it disfigures and stretches the original image. On the other hand, microlensing neither creates multiple images nor distorts the original image, instead, it causes the incoming light flux to vary periodically.

Due to their compact nature, PBHs could serve as foreground objects for microlensing and by measuring the rate of change of the intensity of the electromagnetic wave emitted by the source, bounds can be set up on the PBH dark matter fraction for a particular mass range.

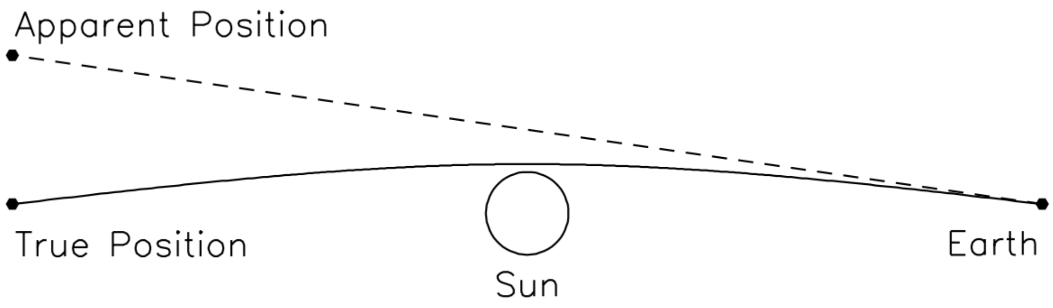

Figure 12: Figure shows a schematic of gravitational lensing. Here, the sun acts as a gravitational lens and bends the electromagnetic radiation that is being emitted by a distant source in the background. Due to this, the apparent position of the object gets shifted, as seen from the earth [105].

Microlensing has been able to provide bounds on f_{PBH} for low mass PBHs, whose masses fall in the range $10^{-7}M_{\odot} < M_{PBH} < 10M_{\odot}$ via the observation of stars inside the Large Magellanic Clouds and Small Magellanic Clouds [89]. The microlensing observations by HSC [90, 91], huge galaxy cluster Icarus (I) [94] and Optical Gravitational Lensing Experiment (Ogle) [95] have provided very strong bounds on PBH dark matter fraction for a vast mass range, $10^{-10}M_{\odot} < M_{PBH} < 10M_{\odot}$. A rather less significant bound for PBHs having masses $M_{PBH} > 10^{-2}M_{\odot}$ comes from the lensing measurements of type Ia supernovae, which restricts $f_{PBH} \leq 0.4$ [96].

5. **Other Constraints**

Other major bounds on the PBH dark matter fraction come from multiple observations including Lyα Forest that detects the distribution of matter at small scales (compared to the cosmological horizon) using the enhancement in the power spectrum due to PBH short noise [97, 98]. As already mentioned, the bounds from accretion largely depend on the type of mechanism one chooses to describe the PBH accretion. Nonetheless, Reference [81] has shown that if PBHs do not constitute the entire dark matter density, accretion onto the PBHs gets amplified due to the presence of non-PBH dark matter halo. Very high energy EM radiation from PBHs may have a noticeable impact on the hyperfine structure of the 21cm H-line. Both accretion radiation and Hawking radiation from PBHs could provide us significant bounds on f_{PBH} by studying their impact on the 21cm H-line [99]. Figure 13 summarizes all the important bounds on the present PBH dark matter fraction for a large mass range.

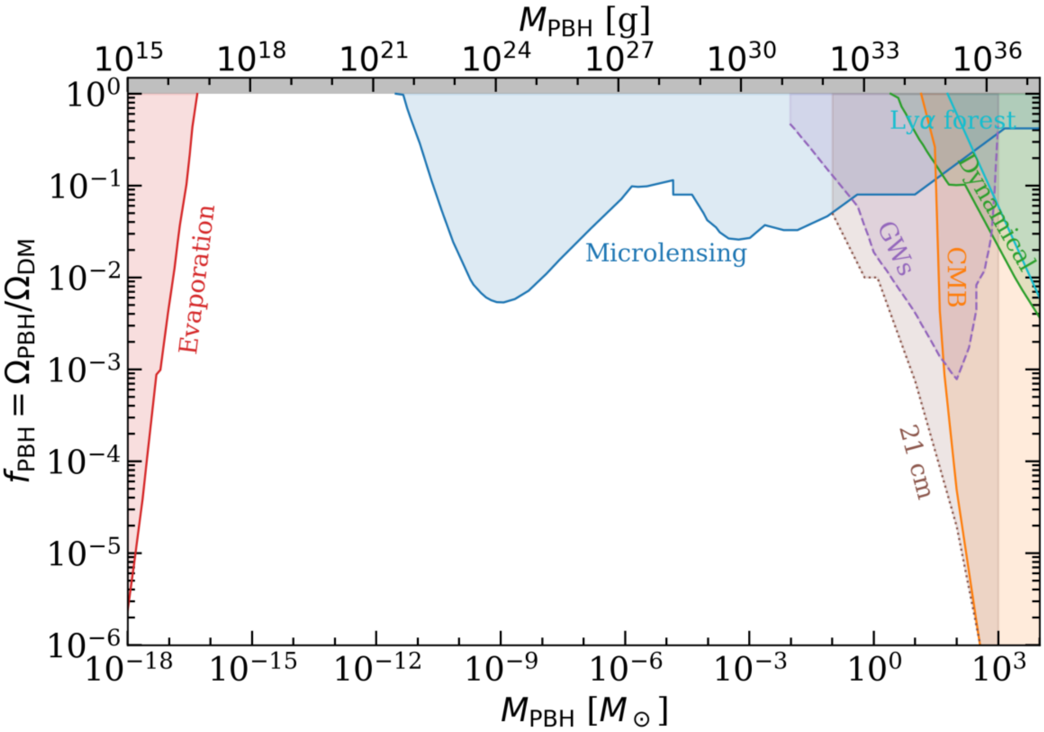

Figure 13: The plot compiles all the major constraints on the dark matter fraction of the PBHs (represented on Y axis) with respect to the mass of the PBHs (represented on X axis in terms of solar mass)[78]. Certain mass windows have still not disregarded the possibility of PBHs making up a considerable fraction of present dark matter.

10 Summary and Future Scope

In the first section of this report, we portrayed a picture of the geometry of the universe and its evolution with cosmic time/redshift from the perspective of Einstein's Theory of General Relativity. Under the assumption that the universe is homogeneous and isotropic on the horizon scales and is filled up with a perfect fluid, we obtained one of the earliest (and one of the fewest as well) analytical solutions to the Einstein Field Equations, famously known as the Friedmann equations, which imply that the universe does not stay the same with time as was previously believed.

In the succeeding sections, we briefly explained the three biggest shortcomings of the standard Big Bang model viz. the flatness problem, horizon problem and the monopole problem. We also summarized how the theory of Cosmic Inflation gives a plausible solution to these problems by introducing a period of vigorous expansion for an extremely short time. Thereafter, we introduced the notion of the Primordial Black Holes (PBHs), which are the central theme of this report, in section 4. We also emphasized why cosmologists have been paying heed to PBHs recently. As PBHs have existed before the matter-dominated era, they are the natural candidates for the non-baryonic dark matter. A very peculiar characteristic of PBHs that makes them even more exciting than stellar BHs is their large range of masses depending on the amplitude of the primordial power spectrum.

With a primary motive to understand the effects of astrophysical processes on the mass and population of the PBHs after their formation, Starting with the accretion phenomena, we summarized the works of Bondi, Hoyle, Lyttleton et al. to get ourselves familiar with the mechanism of accretion onto compact body such as a black hole. We then computed the accretion rates for Hoyle-Lyttleton accretion as well as Bondi-Hoyle accretion. Bondi-Hoyle accretion being much more realistic, its formulas were then applied in section 6 to explore the accretion onto the PBHs. It is necessary to convince the reader that accretion indeed plays a major role in increasing the PBH mass throughout cosmic history so, we gave a rough estimate of the increment in PBH mass due to accretion solely to prove our point, Of course, the PBHs do not gain much mass during the radiation dominated era but as we enter the matter-dominated era, accretion could amplify the PBH mass almost by the order two. Next, we derived the accretion rate for the case of individually accreting PBHs. The viscosity effects of Hubble friction and Compton scattering due to CMB-electron coupling in the early universe lead to the reduction in the rate of accretion, which can be seen from the Equation (97). Similarly, we computed the accretion rate for the case of a PBH+halo system and discussed the enhancement in the amount of accretion due to the extended cloud of dark matter halo.

We then, shifted our attention to the formation of PBH binaries, a topic that has gained enormous popularity even since LIGO-VIRGO detected the GWs from the binary BH mergers [82]. Afterwards, we studied separately the PBH formation mechanisms in the early universe as well as in the late universe. Simultaneously, we calculated the merger rates for both the cases and concluded using Figure 10 that the contribution by the early time PBH binary mergers to the LIGO-VIRGO observations should be much more than the late time PBH binary mergers.

In the following section, accretion onto the PBHs was considered after the formation of binaries. The condition that determines which of the global accretion or local accretion would dominate was also discussed along with the computation of the accretion of the binary as a whole. In the end, Equation (156) revealed a very interesting property of the PBH binary accretion; the accreting binaries end up balancing each other's masses resulting in a kind of equilibrium condition. Finally, we concluded our report by discussing the major bounds on the fraction of dark matter that PBHs could consist of (f_{PBH}). Observations coming from different missions such as LIGO-VIRGO, PLANCK, EUCLID, MACHO, OGLE, Subaru-HSC etc have jointly established numerous constraints on the PBH dark matter fraction as a function of their masses. In the end, it was noticed that there are four chief mass windows where f_{PBH} could still be 'non-negligible'. These mass windows are

1. $10^{-17}M_\odot < M_{PBH} < 10^{-11}M_\odot$
2. $M_{PBH} \approx 10^{-5}M_\odot$
3. $M_\odot < M_{PBH} < 10M_\odot$
4. $10^{14}M_\odot < M_{PBH} < 10^{18}M_\odot$

If one looks at the citations per year of Carr and Hawking's classical paper on PBHs [22] it would be clear that after 2016, the citations have inflated tremendously every year. This suggests that not only the cosmology community but scientists from other physics backgrounds have also started taking PBHs seriously. Hence, a profound growth in the PBH research has been observed over the past few years. Still there lie many unanswered questions and debated conclusions that may become the main areas of interest in PBH cosmology. The best example would be *Primordial non-Gaussianity.* A majority of works on PBH abundance calculations for a given inflationary potential assume a Gaussian PDF. Now, such an assumption is valid as long as the Linear Perturbation Theory remains valid, which is not the case for the power spectra, where the peak of the amplitude at $10^{-4} - 10^{-2}$ orders at small scales. Therefore, the corrections due to non-Gaussian statistics are very important to fully understand the production of PBHs in the early universe. Moreover, it was recently pointed out by Reference [100] that the slow roll condition is violated between the CMB scales and the scales where PBHs are produced. Thus, it is not feasible to compute the curvature power spectrum using the slow-roll approximation. Instead, the power spectrum needs to be calculated by numerically solving the Mukhanov-Sasaki equation. These issues will be discussed in more detail in future works.

A Press-Schechter Formalism

The Press-Schechter formalism, developed in 1974 by William Press and Paul Schechter [64], is one of the most popular methods to compute the population density of galaxies, halos or compact objects as a function of their mass. The concept of mass function is extremely important and has profound applications in a variety of areas in cosmology such as calculating the initial abundance of PBHs, galaxy surveys, formation of large scale structures etc. Despite its simplistic and doubtful assumptions, the results of PS formalism agree well with numerical simulations.

The central idea of the PS formalism is that density fluctuations above a certain threshold (δ_c) result in the creation of an object. The calculation begins with the assumption that the collapsing region of a certain volume where the density contrast is higher than the critical density contrast (δ_c) is directly related to the mass of region ($> M$). We denote the probability of such collapse by $P(> M)$

$$P(> M) = P(\delta > \delta_c|M) \tag{175}$$

The density contrast is smoothed over a region R corresponding to the mass M and denoted by $\delta_{sc}(x)$. This contrast has a Gaussian distribution and its variance is denoted by σ_{sc}. The probability can be calculated by

$$P(> M) = P(\sigma_{sc} > \sigma|M) = \frac{1}{\sigma(M)\sqrt{2\pi}} \int_{\delta_c}^{\infty} \exp\left\{\frac{-\delta^2}{2\sigma^2(M)}\right\} d\delta \tag{176}$$

This expression can be rewritten in terms of an error function such that

$$P(> M) = 1/2 \left\{1 - \mathrm{erf}\left(\frac{\nu}{\sqrt{2}}\right)\right\} \tag{177}$$

where $\nu = \frac{\delta_c}{\sigma(M)}$ is the relative density threshold. The formula (177) undercounts the probability of the collapse by a factor 2 and hence, it has to be multiplied by 2

$$P(> M) = \left\{1 - \mathrm{erf}\left(\frac{\nu}{\sqrt{2}}\right)\right\} \tag{178}$$

The probability distribution equation (178) directly tells us about the fraction of the universe's mass that is being collapsed. The quantity $\frac{\partial P}{\partial M} dM$ can be thought of as a fraction of the universe's mass inside the collapsing dark matter halos whose masses range from $M - dM/2$ to $M + dM/2$. An explicit relation can be found between the mass function of this dark matter halo ($n(M)dM$) and $\frac{\partial P}{\partial M} dM$, again for the mass range $(M - dM/2, M + dM/2)$, such that

$$n(M) = \frac{\rho_M \partial P}{M \partial M} = \frac{\rho_M}{M} \frac{\partial}{\partial M} \left\{1 - \mathrm{erf}\left(\frac{\nu}{\sqrt{2}}\right)\right\} \tag{179}$$

After a little simplification, the expression for the dark halo mass function becomes

$$n(M) = -\sqrt{\frac{2}{\pi}}\frac{\rho_M}{M}\frac{\delta_c \partial\sigma}{\sigma^2(M)\partial M}\exp\{-x^2\} \tag{180}$$

where $M/\rho_M n(M)dM$ denotes the mass fraction associated with the dark matter halos having masses in the range $(M - dM/2, M + dM/2)$. As already said, the distribution of the density contrast is assumed to be Gaussian which defines the power spectrum of the density fluctuations $\mathcal{P}(k)$. The variance of the density contrast depends on the power spectrum. This results in the following simplified form of the Equation (180)

$$n(M) = \sqrt{\frac{2}{\pi}}\frac{\rho_M}{M^2}\left|\frac{\partial \log\sigma}{\partial \log M}\right|\exp\{-\nu^2/2\}\nu \tag{181}$$

Equation (181) is the mass function of the dark matter halo calculated using the Press Schechter formalism.

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